

Security Implementation for Guarantee of Payment System

Cambodia University of Technology and Science Faculty of Engineering

By **Reaksmey Eb** (ID: 6023010017)

Supervisor: Dr. Engelking Elias Jacob Mentor: Mr. Long Makara

September 2026 Phnom Penh, Cambodia

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of **Bachelor's Degree in Cybersecurity**

Declaration of Originality

I, Reaksmey Eb, hereby declare that this thesis, titled *Security Implementation for Guarantee of Payment System*, submitted in partial fulfillment of the requirements for the degree of Bachelor's Degree in Cybersecurity at the Cambodia University of Technology and Science, is my own original work. This work was carried out during my industrial placement at Intercare Hospital from 5 March 2026 to 4 September 2026. All sources and references used were properly acknowledged and cited.

Full name: Reaksmey Eb Date: 1 September 2026

Report Approval

This report, titled *Security Implementation for Guarantee of Payment System*, submitted by Reaksmey Eb (ID: 6023010017), has been approved by the following members of the examination committee as a partial fulfillment of the requirements for the degree of Bachelor's Degree in Cybersecurity at the Cambodia University of Technology and Science.

Report Submission Approval

Supervisor: Dr. Engelking Elias Jacob, Head of Digital Health, Intercare Hospital Mentor: Mr. Long Makara, Lecturer, CamTech University

Examination Committee

- Committee 01: [Name] (Title, Institution)
- Committee 02: [Name] (Title, Institution)
- Committee 03: [Name] (Title, Institution)
- Committee 04: [Name] (Title, Institution)
- Committee Chair: [Name] (Title, Institution)

Acknowledged by Name, Title: Date of Approval:

Acknowledgements

My warm thanks go to CamTech University for the opportunity to undertake this placement, and for its support throughout, from the very start of the process to its completion.

I am genuinely thankful to Intercare Hospital for hosting this placement within such a professional and welcoming environment.

My deepest gratitude to my supervisor, Dr. Elias. You placed your trust in me even though I had no real experience and lacked the skills this work demanded. You stood by me through every situation, followed up

when I needed it, and listened patiently even on my hardest days, always ready to talk things through with me. You pushed me further than I ever thought I could go, and helped me grow into someone I wasn't six months ago. This time under your guidance has been truly precious to me.

I am profoundly grateful to my mentor, Loukru Makara. Your steady direction, especially at moments when I was uncertain how to proceed, was invaluable throughout this project. I appreciate the patience and encouragement you offered at every stage.

I am grateful to Dr. Ebner, Deputy CEO of Intercare, whose contribution made this project possible to carry through in many ways.

Special appreciation goes to the IT team, Bong Udom, Bong Chanthra, and Bong Savda, who formed the backbone of this project's technical work, offering ideas, consultation, and support at every step.

I am truly thankful to the Surgical Coordinator team, Mrs. Debbie and Mrs. Riza; the Finance team, Bong Kimheng; and the Doctor team, Dr. Denny, Dr. Justyna, and Dr. Visoth, for your support and the time you spent answering our endless questions along the way.

To my dear friends on the GOP-MVP project, Rith (Soputhearith), who led development and communication, and Chamrath, who was responsible for data organisation and the ANZER adaptation, I am grateful for your collaboration and dedication throughout this project. This project would not have succeeded without the work each of you contributed, and I value the effort we put in together.

Lastly, I am immensely grateful to my family for your continued support throughout this placement. Your patience and encouragement, even without always knowing the details of the work, meant more than you may realise.

Finally, my genuine appreciation goes to everyone else who contributed to this project in ways not individually named here. Your assistance is sincerely appreciated.

Abstract

Intercare Hospital processes insurance Guarantee of Payment (GOP) requests by hand, which causes delays, errors, and inconsistency. Automating this work places patient clinical data and hospital pricing in a single system used by five staff roles with different responsibilities. The access rules applied to that system were not derived from security frameworks alone; they were developed through a socio-technical process, translating organisational responsibility, gathered through structured, repeated engagement with clinical, financial, and administrative stakeholders, into enforceable software rules. Selected security controls were implemented and assessed for that system, covering identity management, access control, audit logging, and vulnerability assessment, with remaining risks documented rather than resolved. The NIST Cybersecurity Framework 2.0 structured the work, with STRIDE used for threat modelling, the 2021 edition of the OWASP Top 10 for code review, and a separate assessment against the OWASP Application Security Verification Standard (ASVS) 4.0.3 at Level 2. Access control combined role-based (RBAC) and attribute-based (ABAC) rules across forty-two defined actions, supported by an append-only audit log. Verification used both automated scanning and manual role-by-role testing of the running system. Static analysis with Semgrep covered 356 files and produced 15 findings. None concerned access control. Manual role-by-role testing of the same system in the same period produced 47 tracked findings, of which 23 were access-control findings. Most were fixed before deployment, and the rest were recorded as deferred decisions. The ASVS assessment separately produced 15 further findings, of which 10 were resolved. Permissions were found to depend on three factors together: the user's role, their assignment to a case, and how far that case had progressed through the workflow. These findings indicate that automated scanning alone cannot verify authorisation logic, because such rules depend on organisational policy rather than on code structure. They also show that access control in clinical workflows must depend on the state of a case, not on the user's role alone. The placement itself demonstrated that this kind of access control depends as much on understanding organisational responsibility as on security engineering.

Keywords: access control, role-based access control (RBAC), attribute-based access control (ABAC), Keycloak, healthcare information systems, security testing

Table of Contents

1. Introduction	11	1.1
Background of the Study	11	1.2
Company Overview	11	1.3
Project Overview	12	1.4
Research Objectives	12	1.5
Thesis Structure	13	
2. Problem Statement	13	2.1
Background of the Problem	13	2.2
Impact of the Problem	14	2.3
Scope of the Problem	15	2.4
Summary of the Problem	15	
3. Literature Review	16	3.1
Access Control Models: RBAC and ABAC	16	3.2
Security Standards and Frameworks	17	3.3
Threat Modelling Approaches	18	3.4
Automated and Manual Security Verification	18	
4. Methodology	19	4.1
Research Design	19	
Stakeholder-Driven Security Design and Appreciative Inquiry	20	4.2
Frameworks and Standards Applied	21	4.3
Threat Modelling	22	4.4
Tools and Technology	23	
5. Implementation	24	5.1
Identity and Access Management	25	5.1.1
Sign-in and Session Handling	25	5.1.2
Account Status and AD Federation	26	5.2
Access Control Model	26	5.3
Audit Logging	28	5.4
Deployment	29	
6. Results and Discussion	30	6.1
Results	30	6.1.1
How the System Was Checked	30	6.1.2
Package and Container Scanning	30	6.1.3
Checking the Project's Own Code	31	6.1.4
Manual Role-by-Role Testing	32	6.1.4.1
What Was Found	32	6.1.4.2
What Was Done About It	33	6.1.5
OWASP Top Ten Assessment	35	6.1.6
ASVS 4.0.3 Assessment	36	6.1.7
How the Finding Counts Relate	38	6.2
Discussion	39	6.2.1
Permission Depends on More Than the Role	39	6.2.2
Why the Scanners Found No Access-Control Problems	40	6.2.3
Layered Checks, and the Guard That Never Ran	41	6.2.4
What This Suggests for Similar Systems	42	
7. Challenges, Conclusion and Future Work	42	7.1
Challenges	42	7.1.1
Technical Challenges	42	7.1.2
Stakeholder Challenges	43	7.1.3
Personal and Organisational Reflection	44	7.2
Conclusion	46	7.3
Future Work	47	
References	51	

List of Tables

- Table 1: Frameworks and standards used, and how far each was applied ... 21
- Table 2: STRIDE categories and how each was addressed ... 22
- Table 3: Tools used in the system and in its verification ... 24
- Table 4: The five roles and what each is responsible for ... 27
- Table 5: Scanning targets and findings ... 31
- Table 6: What Semgrep found ... 32
- Table 7: The 47 findings by type ... 32
- Table 8: The six kinds of access-control finding ... 33
- Table 9: Examples of work left for later, and why ... 34
- Table 10: OWASP Top 10:2021 ratings ... 36
- Table 11: ASVS 4.0.3 results by chapter ... 37
- Table 12: Summary of manual-testing findings ... 38
- Table 13: The three factors, with an example of each ... 39
- Table 14: Work remaining, by priority ... 47

List of Figures

- Figure 1: The GOP process on paper compared with the process in the system ... 14
- Figure 2: The seven phases of the security work ... 20
- Figure 3: The Appreciative Inquiry cycle used in weekly team meetings ... 20
- Figure 4: The sign-in path ... 25
- Figure 5: Account status after AD federation ... 26
- Figure 6: The three factors evaluated in every access decision ... 28

1. Introduction

This chapter introduces the setting of the project and the problem it addresses. It describes how Guarantee of Payment requests were handled at Intercare Hospital before the system was built. It then gives an overview of the hospital and the project, states the objectives of this thesis, and outlines the structure of the chapters that follow.

1.1 Background of the Study

Hospitals and insurance companies work together through a process known as Guarantee of Payment (GOP). Before a planned admission, the hospital asks the insurer to confirm in writing that it will cover the cost of treatment, so the patient can be treated without paying up front. The request is not a single document but a package: patient details, a clinical description of the planned procedure, statements from the treating doctors, and an itemised cost estimate. Because insurers set their own forms and required fields, assembling that package is manual work that falls on hospital staff, and the effort grows with the number of insurers a hospital works with.

At Intercare Hospital in Phnom Penh, this process was carried out manually, and a single request could pass through four departments before it reached the insurer. The result was slow, inconsistent, and tiring for the people doing it. Chapter 2 describes that process in detail.

Automating this process removes much of that effort, but it also changes where the risk sits. Information that once lived on paper in a single office now sits in a database that many people can reach across the hospital network. Decisions that were previously made in person, such as who is allowed to confirm a cost or approve a clinical statement, become rules written into software. A GOP request contains both clinical information about a patient and a financial commitment on behalf of the hospital, so a mistake in those rules is not only a privacy problem but a financial one.

Security therefore could not be treated as something added after the system worked. It covers more than one question: who is allowed into the system, what each person may do once inside, what record is kept of what they did, and whether the software and servers the system runs on are themselves safe. These questions had to be answered alongside the workflow rather than after it. This thesis describes the requirements that were

identified, the security that was designed and built to meet them, and how the result was tested and revised.

1.2 Company Overview

Intercare Hospital is a private healthcare provider in Phnom Penh, Cambodia, based at the Olympia Medical Hub Building on Street 161, Veal Vong, 7 Makara. The hospital's stated aim is to become the most trusted healthcare provider in Cambodia, with values organised under a framework called I-CARE, covering innovation, compassion, accountability, responsiveness and empowerment [1].

The hospital opened in 2020 as a general practice clinic and has added services each year since: laboratory and imaging in 2021, maternity, inpatient and emergency care in 2022, and operating theatre services, physiotherapy and a visa health check centre in 2023. Paediatrics, a child wellbeing centre and a set of community clinics under the Intercare Express name followed, with a medical centre in Siem Reap planned [1].

The hospital employs more than two hundred medical professionals, Cambodian and international, representing over twenty nationalities, and treats Cambodian families, expatriates, company employees and travellers. The hospital's services are delivered in partnership with more than forty insurance providers, which makes the Guarantee of Payment process a routine part of daily work rather than an occasional task [1].

1.3 Project Overview

This project was carried out within the hospital's Digital Health department, which is responsible for the systems that support clinical and administrative work. The aim was to replace the manual cost estimation and Guarantee of Payment process with a web-based system, so that a request could be prepared, reviewed and sent from one place instead of being assembled by hand across several departments.

The wider effort is planned in three phases. This project delivers Phase 1, built over six months as the internal foundation that later phases can extend without being rebuilt. Phase 1 covers the hospital's own side of the workflow: creating a case from patient and encounter data, building a cost estimate, generating the GOP request and its document output, tracking its status through approval, rejection or revision, and keeping an audit trail. The system does not connect to insurer systems. Finance reviews each case and sends the request from within the system; nothing goes out automatically. The insurer's reply arrives in a staff Outlook mailbox rather than in the system, so Finance records the answer by hand, and a case needing revision is corrected and sent again by the same path. Claims generation and submission belong to later phases. The system's data model follows the HL7 FHIR standard, Fast Healthcare Interoperability Resources (FHIR), which shaped several of the security decisions described later. At the time of writing, some parts of this workflow were complete and working, while others were still being finished.

The system was built by a team of three students on industry placement, each from a different major: software engineering, data science and artificial intelligence, and cybersecurity. My major is cybersecurity, and I was responsible for the security of the system. That covered working out what needed to be protected, designing and building the controls to protect it, and testing whether they held. This thesis describes that security work.

The system was built for internal use within the hospital rather than for public access, and runs on the hospital's own servers, not in the cloud. This means no outside cloud provider shares any responsibility for security. The hospital alone is responsible for protecting the system and the server it runs on. This shaped much of the security design.

1.4 Research Objectives

The general objective of this work was to build security into the Guarantee of Payment system as it was developed, rather than adding it once the system was working. The specific objectives were as follows.

1. To identify what the system needs to protect, and to establish the security requirements of the workflow through structured engagement with the hospital's clinical, financial, and administrative stakeholders, including the separations between roles that already exist in hospital practice.
2. To select security standards suitable for a small internal health system that can also be extended as the system grows, and to state how far each one is applied.

3. To design and implement the security of the system: signing in, what each user may do, how their actions are recorded, and the security of the software the system runs on.
4. To evaluate the security before deployment, using manual testing, automated scanning, and a separate assessment against the OWASP Application Security Verification Standard, and to record what was found.
5. To compare what each method detected, and to discuss what that means for verifying access control in systems of this kind.

1.5 Thesis Structure

This thesis is organised into seven chapters.

- Chapter 1 introduces the setting, the hospital, the project, and the objectives of this work.
 - Chapter 2 describes the problem: how Guarantee of Payment requests were handled manually, what that cost the hospital, and which part of it this project addresses.
 - Chapter 3 reviews existing work on access control models, security standards for health information systems, threat modelling, and how access control is verified.
 - Chapter 4 sets out the method, including the stakeholder-driven process used to understand the workflow, the frameworks applied, and how far each one was used.
 - Chapter 5 describes what was built: how users are identified, the access control model, the audit trail, and the security of the deployment.
 - Chapter 6 reports what the verification produced and discusses what those results mean.
 - Chapter 7 covers the difficulties encountered, a reflection on what the placement changed personally, for the team, and for the hospital, the conclusions drawn, and the work that remains.
-

2. Problem Statement

This chapter sets out the problem the project addresses. It describes how Guarantee of Payment requests were handled before the system was built, what that process cost the hospital, which part of it this thesis is concerned with, and what the problem amounts to when stated in short.

2.1 Background of the Problem

A Guarantee of Payment request is assembled from parts held by different people. The patient's identity and insurance details sit with the front desk, the clinical description comes from the surgeon, a separate assessment comes from the anaesthetist, and Finance builds the itemised cost from the hospital's own pricing. Each insurer sets its own form and required fields, so the same case may be presented differently depending on who is being asked to pay.

At Intercare Hospital this was done on paper, and the paper travelled. The surgeon and anaesthetist completed their parts and handed the file to the Surgical Coordinator, who checked it for missing information. Where something was missing, the Coordinator went back to the doctor concerned, who either supplied it themselves or gave permission for the Coordinator to enter it on their behalf. Once complete, the file went to Finance for costing, returned to the Coordinator for the patient's acknowledgement, and if the patient agreed went back to Finance to be sent to the insurer. If the patient declined, the case ended there.

The paper file was the control. Only the person holding it could act in that case. Nothing moved forward unless someone carried it, and an incomplete case went back to the doctor. The order of the work, the point at which each group's part ended, and the separation between the clinical and financial sides were all carried by the movement of that paper. None of it was written down as a rule, because none of it needed to be.

Automating this is not just moving the same process onto a screen. Once the file becomes a database record, everyone can reach it at any time, and the sequence the paper enforced no longer exists.

The new system removes the Coordinator as a required stop. The doctors now pass the case to Finance directly. The Coordinator now only watches progress and sets case priority; editing another role's part of a case is not something the Coordinator does.

Figure 1: The GOP process on paper compared with the process in the system. Amber marks where the two differ. A patient who declines ends the case; where the insurer asks for changes, Finance revises the request

and sends it again.

Version 1 (on paper, Coordinator takes action): Surgeon (starts and completes) → Anaesthetist (continues the case) → Coordinator (checks the case) → Finance (builds the cost estimate) → Coordinator (sends to the patient) → Patient (accepts or declines) → Finance (mails the insurer) → Insurer (three answers)

Version 2 (in the system, Coordinator steps out of the flow): Surgeon (starts and completes) → Anaesthetist (submits the case) → Finance (checks the case, builds the cost estimate, and sends it to the patient) → Patient (accepts or declines) → Finance (mails the insurer) → Insurer (three answers)
With Coordinator shown separately as: "no longer a required step; monitors progress only"

On paper, the Coordinator had to walk over and ask before changing a doctor's form. In software, nothing physical stops anyone, so each rule the paper enforced had to be rebuilt as something the code actively checks. The surgeon and anaesthetist can no longer edit a case once it reaches Finance; the system simply refuses the change. Every rule the paper enforced is now enforced by code, rather than by the physical act of having to walk over and ask.

2.2 Impact of the Problem

The manual process cost staff time. Preparing one request meant collecting information that already existed elsewhere. Staff walked between departments to get it, then typed the same patient details more than once as the file moved. A request passed through four departments before it left the hospital, and each hand-off waited on someone being free to receive it.

That time cost was worst at one specific point: the handover to Finance. The surgeon and anaesthetist wrote their parts by hand, and Finance then re-entered all of it digitally, because both the patient consent form and the insurer's request had to be produced as documents. With each insurer setting its own format, the same handwritten case could be typed out in more than one layout.

Underneath that delay sat a harder problem: the process depended on people who are hard to reach. Finance and the Coordinator could not move a case forward until the surgeon and anaesthetist had entered their parts, and staff report that those parts were sometimes forgotten or supplied late. Chasing a doctor who is in theatre or between patients is not quick, so a case could sit waiting on one missing figure.

The same paper that could not be chased quickly also gave staff no way to tell which case needed attention first. With cases held as paper files, there was nothing to show which had been waiting longest or which needed attention first. Staff report that a case opened early could finish last, and that an emergency case could be delayed simply because the file was hard to find among the others. Nothing about a stack of paper tells you what is urgent.

All of this repeated handling had a further cost beyond time: accuracy. Details typed a second or third time can be typed differently, a date entered as the fourth month rather than the third, for instance. The information the doctors supplied was correct; the copying was where it went wrong. A mismatch between the patient record, the clinical description and the cost is the kind of error an insurer returns rather than resolves. And because each insurer sets its own form, a missing field found late meant the request came back and the work was done again.

One thing the paper did do well, almost without meaning to, was keep a record of all this friction. The file showed who the doctors were, who handed it on, and when. That record existed as a by-product of being paper: a signature had to be written, a date entered, and a correction crossed out in view of whoever held the file next. Software keeps none of that by itself. A database shows the current value of a field, not who set it or what it was before. Replacing the paper meant that this record, too, had to be built deliberately, or there would be none.

2.3 Scope of the Problem

Not all of the problems described above are addressed here. The security work covers signing in, what each user may do, how their actions are recorded, and the security of the software the system runs on. It also covers the administration screens themselves: the interface for enabling and disabling accounts, and the view onto the audit trail. The application logic, the integration with the hospital information system, and the

database design were the work of my teammates, and appear in this thesis only where a security decision depends on them.

Because the system is not internet-facing, this phase focused primarily on authenticated internal users, compromised internal accounts, and misuse of legitimate access, while recognising that external compromise could still occur indirectly through the hospital network, for instance through a stolen credential, a compromised endpoint, or lateral movement from elsewhere on the network. Two checks were outstanding at the time this scope was set: a penetration test, carried out from the hospital's side, combining the hospital's own IT security team with Canada Bank's IT security team, and a review of the container settings against the Center for Internet Security (CIS) Docker Benchmark. Both have since been carried out, though it is not confirmed whether the container review specifically followed the CIS Benchmark, or used a different method. Going live also required a signed non-disclosure agreement, hospital approval, and privacy safeguards, all agreed at the start of the project.

2.4 Summary of the Problem

The manual Guarantee of Payment process was slow, repetitive and open to error. It did keep a record of who did what, but only because the paper carried signatures, dates and visible corrections. Automating it removes the costs. It also removes two things that came free with the paper. The first is the physical route that quietly enforced the order of the work: who could act on a case, at what point their part ended, and where the clinical side stopped and the financial side began. The second is the record itself.

Software will not make people work in order, and it will not keep a history. You have to build both. Every rule must be written into the code. Every action must be saved on purpose. On paper, someone nearby might spot a mistake. In software, no one is nearby. If a rule is wrong, nothing catches it.

The question is not whether the process can be automated. It is whether the system can be trusted to let the right people do the right things. Answering it means listing every action the system allows, deciding the rule for each one, putting the check somewhere the user cannot get past, recording what happened, and then testing that it works.

3. Literature Review

This chapter reviews the existing work this project uses. The work covered the security of the system as a whole: signing in, what each user may do, how their actions are recorded, and the security of the software the system runs on. A systematic review of access control in electronic health record systems groups this field into four categories, discussed in section 3.1, and those categories match the strands of work described in Chapter 5.

The review is weighted towards authorisation, because that is what most of this project's findings are about. Section 3.1 covers the models of authorisation and what each can express. Section 3.2 covers the standards and frameworks applied, both health-specific and general, which between them address authentication, transport and audit. Section 3.3 covers threat modelling. Section 3.4 covers how access control is verified, which is where the main result of this project lies.

3.1 Access Control Models: RBAC and ABAC

Access control decides which users may do what, and to which records. Two models are treated as the classic approaches, and FHIR's security guidance describes both for health systems [2]. In role-based access control (RBAC), permissions are grouped into roles, each role describes a job function, and users are given roles. If a user's role holds the permission, access is granted [2]. Permissions follow the job rather than the person, which suits a hospital, where roles such as surgeon, anaesthetist and finance officer already exist before any software is written.

In attribute-based access control (ABAC), the decision is made from attributes and conditions rather than from roles alone. The National Institute of Standards and Technology (NIST) describes ABAC as a method in which authorisation is determined by evaluating attributes of the subject, attributes of the object, the requested operation, and in some cases the conditions of the environment, against rules that describe which operations a given set of attributes allows [3]. NIST places ABAC on a spectrum of logical access control that

also includes access control lists and role-based access control, so it is a development of earlier approaches rather than a replacement for them [3]. The practical gain is that rules may refer to the record itself. "A surgeon may create a case" is a role rule. "A surgeon may edit this case" is not, because the answer depends on the case, who it is assigned to, not on the surgeon.

Both standards recognise that a decision may also depend on the circumstances of the request, and both make room for it in their models. NIST calls this environment conditions and lists it as one of the four inputs to a decision, alongside rules, subject attributes and object attributes [3]. FHIR uses the term transaction context and names, among its examples, the workflow state of the request [2]. The workflow position of a case is therefore recognised in both, though under different names and in different parts of the model.

Work on access control in health systems has followed the same direction. A systematic review of twenty studies on electronic health record systems found attribute-based access control to be the most common authorisation mechanism, and grouped the mechanisms proposed into identification, authentication, authorisation and accountability [11]. The review also names what the literature has not resolved: multi-factor authentication, emergency access, patient consent and accountability are identified as gaps, with accountability addressed by only two of the twenty studies. Others have combined the two models directly. Atlam and Yang propose a healthcare model integrating role-based categorisation with attribute-based control over user attributes and environmental factors, observing that models which allocate permissions by predetermined roles or policies adapt poorly to the changing circumstances of a working healthcare system [12].

The three inputs are not alternatives. Roles carry the decisions that follow the job, attributes carry those that depend on the individual record, and the case's position in the workflow carries those that depend on how far it has progressed. What neither standard describes is how they combine in practice, what it costs to enforce them consistently, or how such rules can be checked once built. Chapters 5 and 6 report that.

3.2 Security Standards and Frameworks

Health information systems carry data that is sensitive in ways ordinary business systems are not. FHIR's security guidance was the health-specific source used here. It is unusual in addressing implementation directly: it recommends OAuth for authenticating users, states that production data should be sent over TLS, describes how audit events should be recorded, and sets out how a server should respond when access is refused [2].

No Cambodian standard exists for health information security that a hospital can adopt. The nearest domestic guidance is the National Bank of Cambodia's Technology and Cyber Risk Management Guidelines, issued in January 2026 [4]. These apply to banks rather than hospitals, so this system is not subject to them. They are, however, the only Cambodian document setting out detailed security requirements for systems holding sensitive data, and the requirements used here concern the protection of that data rather than financial services.

The guidelines require role-based access control with least privilege, a maintained access control matrix, segregation of duties so that no single person can enter, authorise and complete a transaction, audit logs that record both successful and refused access and cannot be altered, encryption of data in transit and at rest, and vulnerability assessment using both automated tools and manual techniques [4]. They also note that Cambodia's adoption of technology is outpacing its regulatory frameworks, and direct institutions to adopt recognised standards where no local regulation exists [4].

Two general frameworks are relevant alongside FHIR: the NIST Cybersecurity Framework, which organises security work into six functions (Govern, Identify, Protect, Detect, Respond and Recover) [5], and the OWASP Top 10, a list of the ten most common categories of web application security risk. The 2021 edition is the one used in this project [6]. Chapter 4 describes how each was applied.

OWASP also publishes the Application Security Verification Standard (ASVS), a checklist of individual, checkable requirements rather than broad categories. Version 4.0.3, released in 2021, sets out roughly 280 requirements across fourteen chapters, covering areas the Top 10 does not break out on its own, such as session management, error handling, and business logic. It also defines three verification levels, from a basic minimum to the most rigorous, so a system can be checked against a level appropriate to what it handles. A newer version, ASVS 5.0.0, was released in May 2025 and is described by OWASP as its most substantial

revision to date [14]. OWASP updated the Top 10 itself in 2025, reordering its categories [13].

Two standards specific to healthcare were also considered. The Health Insurance Portability and Accountability Act (HIPAA) sets security requirements for healthcare data in the United States [15]. ISO/IEC 27001 sets requirements for an organisation-wide information security management system. Its companion, ISO 27799, adapts those requirements to health information specifically. It cannot be used on its own; it still requires ISO 27001 as its foundation [16].

None of these standards states who in a particular hospital may perform a particular action on a particular case. Each addresses authentication, data protection, or general code quality, but none reaches down to a specific organisation's division of responsibility. Those rules have to be drawn from the workflow itself, which is where Chapter 4 begins.

3.3 Threat Modelling Approaches

Threat modelling is the practice of working out what could go wrong with a system, without testing it. It asks what someone might attempt against each part of a system, and what would follow if they succeeded. OWASP treats it as a design activity. Its category for insecure design calls for greater use of threat modelling in authentication, access control and business logic, and separates design flaws from implementation defects, on the grounds that a design flaw cannot be fixed by careful coding [6].

STRIDE is the most widely used method. It groups threats into six kinds [7]: spoofing, where someone acts as another user; tampering, where data is changed without permission; repudiation, where a user denies an action and no record contradicts them; information disclosure, where data reaches someone who should not see it; denial of service, where the system is made unavailable; and elevation of privilege, where a user gains rights they were not granted. Each part of a system is examined against all six.

The value of the method is coverage, not discovery. Working through six categories for every component makes it harder to miss a whole class of threat. What STRIDE does not do is say which threats matter most, or supply the rules that would prevent them. It can tell you a system may be vulnerable to tampering. It cannot tell you which staff member should lose the ability to edit a record once another department has reviewed it. That judgement comes from the workflow, not from the method.

Threat modelling is usually described as a design-time activity, carried out before a system is built. It can also be applied to a system that already exists. In that case it organises and classifies what is known, rather than predicting what has not yet been found. Chapter 4 describes which of these applies to this project.

3.4 Automated and Manual Security Verification

Writing down who may do what is one task. Checking that the software actually does it is another. A rule table states what should happen, but it says nothing about what the code does. The two can disagree without anyone noticing, because when a system allows something it should have blocked, nothing appears to go wrong. Not every strand of this work presents the same difficulty. Dependency versions, container images and secrets left in a repository can each be checked by tools, because each has a recognisable form. Access control does not, which is why it is the strand this section concentrates on.

Automated tools are the usual first step, and their main difficulty is well documented. Sun, Xu and Su point out that access control rules are specific to each application [8]. Unlike an injection flaw, which looks much the same in any program, there is no general pattern for a tool to search for. The rules have to be supplied, and writing them by hand is slow and tends to produce rules that are missing, incomplete or wrong. Their own tool works around this by inferring the rules from the source code: it builds a list of the pages each role can open, compares those lists to find pages only some roles should reach, and then tests whether other users can reach them anyway.

Inferring the rules from code is where the method reaches its limit. Zhu and colleagues tested six open source applications and found that the assumptions built into earlier tools held those tools back [10]. They tried a different approach, asking developers to supply the rules and turning those answers into patterns the tool could then apply. This found twenty vulnerabilities the automatic methods had missed. A survey published in 2024 reaches the same conclusion, comparing the methods available and listing the problems that remain unsolved [9]. The pattern across all three is that access control rules come from the organisation rather than from the code, and that a tool performs better once a person supplies them.

This is why automated analysis cannot be the whole of the work, and the guidance says as much. OWASP asks developers to write tests for access control [6]. The National Bank of Cambodia's guidelines require both automated tools and manual techniques [4]. Neither explains how to test permissions that change as a case moves from one stage to the next. Chapter 6 reports what each method found when both were used on the same system in the same period.

4. Methodology

This chapter describes how the security work was carried out. It covers how the work was organised, which frameworks and standards were applied and how far, the threat model, and the tools used.

4.1 Research Design

I planned the security work in seven steps: planning and requirements, choosing tools, design, building, testing before deployment, deployment, and checking afterwards. These were grouped under the functions of the NIST Cybersecurity Framework 2.0, and Figure 2 shows them. Five were completed at the time this was written; deployment, carried out by a teammate, had not fully succeeded, and the checks that follow deployment had not begun. Deployment has since been completed, every fix verified locally during this work is now live on the hospital's server, though the post-deployment checks (section 7.3) remain outstanding. The five phases that ran did not run once each in order. Every time I checked something, I found a reason to change a decision I had already made, so the design, the build and the testing each came round more than once. That is why I call the method build-and-check cycles. The seven steps say what I planned; the cycles say how it actually ran.

Figure 2: The seven phases of the security work, mapped to the functions of the NIST Cybersecurity Framework 2.0. Phases 3 to 5 were revisited as checking changed earlier decisions. Phases 6 and 7 were not complete at the time of writing.

Umbrella: NIST Cybersecurity Framework 2.0 - Phase 1: Planning and requirements.

Stakeholders, local hosting constraint, threat model. Frameworks: NIST CSF (Govern, Identify), STRIDE - Phase 2: Research and tool selection. Entra ID trial compared with locally hosted Keycloak. Frameworks: NIST CSF (Identify) - Phase 3: Design. Permissions matrix, combined role and attribute model. Frameworks: NIST SP 800-162, HL7 FHIR, STRIDE, NIST CSF (Protect) - Phase 4: Implementation. Keycloak deployed locally, sign-in, access rules, audit logging. Frameworks: NIST CSF (Protect, Detect) - Phase 5: Pre-deployment testing. Role-by-role testing, automated scanning, OWASP self-assessment. Frameworks: NIST CSF (Detect), OWASP Top 10:2021, NBC guidelines - Phase 6: Deployment. Readiness summary, handover to the development side, known limitations. Frameworks: NIST CSF (Respond) - Phase 7: Post-deployment validation. Container benchmark, formal assessment with IT Security, code review. Frameworks: OWASP Testing Guide, CIS Docker Benchmark, NIST CSF (Recover)

Phases 3 to 5 were revisited as checking changed earlier decisions. Phases 6 and 7, shown dashed, were not complete when this was written.

The work covered four things: signing in, what each user may do, how their actions are recorded, and the security of the software the system runs on. Going round more than once mattered. I could not check the rules until I had written them, and I could not write them correctly until I had tested the system.

I did not begin from a written table. The code already existed, written by my teammate, so I started by reading it and writing down what it allowed each role to do. That gave the first version of the rules document. I then went through it against the workflow the team needed, adding rules and changing others, and writing down every difference I found. Some of those differences I could settle myself. Others needed the team to decide. A few are still open, because they depend on how staff want to work. Differences where the code was simply wrong were listed separately from differences where nobody had decided yet. Both lists are in the rules document. This review was the first of three checks.

The second check was manual testing, role by role, on the running system on my own machine. I signed in with one account for each of the five roles and followed a case from creation through to the insurer's decision. Where the system behaved differently from the rules, I recorded a finding. I then either fixed it, left it with a

written reason, or closed it because the system was right and the rule was wrong.

The third check used automated tools on the same version of the code, in the same period: one for the packages the system depends on, one for the container images, and one for the team's own code and any passwords left in it. I ran the scanners and the manual testing on the same version at the same time, on purpose. It means the two sets of findings can be compared directly, rather than across a version that had changed in between.

A fourth pattern ran alongside these three. I passed changes to the teammate handling deployment, and when a problem turned out to be on the security side, it was reported back to me, I corrected it, and passed it on again. Some problems only appeared once the system was installed, and those reached me through those reports rather than through my own testing.

Each cycle produced documents as well as code changes: the access control rules table, a record of each scan, an assessment against all ten OWASP categories, and a summary of what remained open and who owned it. I wrote these while working, not afterwards, and they are where the numbers in Chapter 6 come from.

I used AI tooling to help write and investigate code, under my direction. What to examine, whether a finding was real, and what to fix or defer were my decisions. I checked what the tooling reported against the running system before accepting it, and several reports turned out to be wrong. The placement taught me that AI can accelerate investigation and implementation, but it does not replace verification. Security decisions remained my responsibility, and outputs were accepted only after they were checked against the code, the running system, and the hospital's operational requirements.

Stakeholder-Driven Security Design and Appreciative Inquiry

The access rules in this thesis were not derived from security frameworks alone. They came from a weekly process the wider team already used at Intercare, following the company's I-C-A-R-E values, Innovation, Compassionate, Accountability, Responsive, and Empowered, through a structured cycle: Define, a relevant Value, Discover, Dream, Design, and Deliver.

Figure 3: The Appreciative Inquiry cycle used in weekly team meetings at Intercare, centred on the company's I-C-A-R-E values. Define and Deliver sit opposite each other in the cycle; Discover, Dream, and Design run between them, with the relevant Value checked at each stage.

Centre: Intercare Values (I-C-A-R-E) - Step 1: Define. Name the topic to report on. Step 2: Value. Choose which of the five values, Innovation, Compassionate, Accountability, Responsive, Empowered, the topic connects to. Step 3: Discover. Give the detail of what was done, the problem, and the open question. Step 4: Dream. Decide what to do next, the ideal outcome. Step 5: Design. Decide how to get there. Step 6: Deliver. Assign who on the team will carry it out.

The cycle repeated every Tuesday morning, and each week's Deliver step fed directly into the next week's Define.

Each Tuesday morning, the three-person team, covering data and adaptation, development and communication, and security, took turns presenting what we had worked on that week, with my supervisor present at every session, and often joined and supported by others from the IT team, the Finance team, the Coordinator team, or the Doctors. For each topic, the person presenting first defined it, then named which of the five values it connected to, before discovering the detail of what had been done, what problem or question had come up, and what remained open. From there, the group discussed what they wanted next, their dream for that piece of work, before moving to how it would actually be designed, and finally who would deliver it.

This process was where the access rules in Chapter 5 were actually decided, not just documented afterwards. Rather than designing permissions from a security framework downward, the team went through each role in the workflow in turn, the surgeon, the anaesthetist, Finance, and the Coordinator, and discussed how each one actually worked on paper before agreeing how the same responsibility should work in the system. The clearest example of this is the shift described in section 2.1: on paper, the Coordinator was a required step between the doctors and Finance. Deciding whether that should carry over into the digital system, or whether the doctors should pass a case to Finance directly, took several weeks of this weekly cycle to settle, since it meant Finance absorbing a step that used to belong to someone else, and every actor needed to understand

and agree with the resulting shape of the workflow before it was built. The rule that resulted, described in section 5.1.2, exists because of that discussion, not because of any security standard.

4.2 Frameworks and Standards Applied

I applied frameworks selectively rather than in full. Table 1 states which version I used and how far each one was applied, including the two that were planned but not carried out in this phase. Where a newer version has been published since this work was done, the version listed is the one that was current at the time, and re-assessment against the newer version is recorded in section 7.3.

Table 1: Frameworks and standards used, and how far each was applied

Framework	Version	Extent of application
NIST Cybersecurity Framework	2.0 (2024)	Six functions used to organise the work; control set not implemented
HL7 FHIR	R5 (guidance cited)	Security section only
NIST SP 800-162	2014	Definition and model of attribute-based access control
OWASP Top 10	2021	All ten categories rated against evidence
NBC Technology and Cyber Risk Management Guidelines	2026	Access control, data security, audit trail and testing sections, used as a benchmark
OWASP ASVS	4.0.3 (2021)	Level 2 target, all 14 chapters assessed, 62 subsections checked
STRIDE	N/A	All six categories
CIS Docker Benchmark	N/A	Not applied; planned for a later phase
OWASP Testing Guide	N/A	Not applied; planned for a later phase

Each source did one job. The NIST Cybersecurity Framework gave the work its structure: the seven phases in Figure 2 follow its six functions [5]. The system's data model already follows FHIR, so FHIR's security section was the obvious place to look for decisions about signing in and encryption [2]. Citing FHIR's current security guidance is a reference to the most recent published guidance available, not a claim that the deployed FHIR server itself implements that same specification release; Table 3 in section 4.4 gives the actual server and version in use. NIST SP 800-162 gave the access control model [3]. The OWASP Top 10 was the list I checked the code against [6]. The National Bank of Cambodia's guidelines gave something to measure the result against, rather than a rule to comply with [4]. The OWASP ASVS gave a second, more granular checklist: a separate pass at Level 2 across all fourteen chapters of the 4.0.3 edition, reported alongside the Top 10 results in Chapter 6 [14]. STRIDE gave the six categories used in the threat model described next [7].

HIPAA and ISO/IEC 27001 were also considered. HIPAA was set aside because it is United States legislation, and this hospital is in Cambodia. ISO 27001, and its healthcare companion ISO 27799, require building a full organisation-wide information security management system. This includes ongoing risk registers, management review cycles, and certification audits. That scope was too large for a single feature built during a six-month placement. NIST CSF was used instead, selectively, as already described.

Two frameworks in the table were not used. The CIS Docker Benchmark checks how containers are set up and run. The OWASP Testing Guide sets out how to carry out a penetration test. Both belong to work that comes after deployment, neither had been done when this was written, and both are listed in section 7.3 as work that remains.

I only used parts of each framework, so I cannot say the system complies with any of them. What I can say is which specific requirements were met and which were not, and each one comes from a document I can point

to. Read Chapter 6 that way, as findings measured against requirements, not as proof of compliance.

4.3 Threat Modelling

I used STRIDE: spoofing, tampering, repudiation, information disclosure, denial of service, and elevation of privilege, to sort the threats this system faces. This was a retrospective threat analysis: I applied it after the system was built, not before. Its job was to file what testing had already found, and to check that no kind of threat had been missed. STRIDE gave the six categories. My findings filled them.

STRIDE suited this project for two practical reasons. The system was small enough that every part could be tested directly, rather than estimated in advance, so no separate scoring step was needed before testing began. The work was also done by one person, not a team, so a method requiring coordination across multiple stakeholders would not have fit.

Five of the six hold defects found during this project. The sixth, denial of service, was not addressed, and is recorded as a limitation.

Table 2: STRIDE categories and how each was addressed

Category	Threat in this system	Status
Spoofing	Someone acts as a user they are not	Fixed; findings detailed in section 6.1.4
Tampering	Case data changed by someone who should not be able to, or at a point when it should be locked	Mostly fixed; one open item remains (section 7.3)
Repudiation	Someone denies an action, and the record cannot prove otherwise	Mostly fixed; one open item remains (section 7.3)
Information disclosure	Data reaches someone who should not see it	Mostly fixed; a field-exposure gap and two unencrypted internal connections remain open (section 7.3)
Denial of service	The system is knocked over so nobody can use it	Not addressed in this phase
Elevation of privilege	A user gains rights they were not granted	Fixed; findings detailed in section 6.1.4

Most of the findings are about people seeing what they should not, or doing what they should not. That is what you would expect from a system where five roles share the same cases. Both kinds are decided by the same three things: the user's role, whether the case is theirs, and how far the case has progressed.

Denial of service was not addressed in this phase. Being reachable only inside the hospital lowers the risk from an outside attacker. It does not remove the risk from a compromised or careless account inside the hospital. Testing for it was not reached in the time available. It is understood to be covered by Canada Bank (IT security department)'s separate review, though those findings are confidential and not reproduced in this thesis. It is recorded in section 7.3.

4.4 Tools and Technology

The tools fall into two groups: those the system is built from, and those used to check it.

Table 3: Tools used in the system and in its verification, current versions at the time of writing

Purpose	Tool	Version
Identity provider	Keycloak	26.7.2
Application framework	Next.js	15.5.24
Session handling	Auth.js (NextAuth)	5.0.0-beta.32

Database	PostgreSQL	16
Database access and migrations	Prisma	7.8.0 (client), 7.10.0 (CLI)
FHIR server	HAPI FHIR	7.4.0
Hospital information system	ANZER, read-only	N/A
Deployment	Docker / Docker Compose	29.7.2 / v5.4.0
Static analysis	Semgrep	N/A
Container image scanning	Trivy	N/A
Dependency scanning	npm audit	npm 11.12.1
Secret scanning	secretlint	13.0.2
Browser-driven testing	Puppeteer	25.3.0

Semgrep and Trivy were run in July 2026. I did not fix which version of each tool was used, so both ran whatever release was current that day.

Keycloak was chosen after an extended trial of Microsoft Entra ID. The hospital needed the system to run on hospital-owned servers, and Entra ID runs only as a cloud service. Keycloak also supports user federation with the hospital's own Active Directory, so staff sign in with the same credentials they already use elsewhere in the hospital, rather than remembering a separate username and password for this system. The application itself never stores or handles user passwords, since sign-in is delegated to Keycloak and, through it, to the hospital's AD. This also matched FHIR's security guidance, which recommends OAuth [2], so the choice fit both the hospital's requirements and the standard the data model follows. Auth.js keeps a user logged in, and the version used is a beta; the makers have not marked it finished, though many production systems use it.

CodeQL and Semgrep both read code looking for mistakes; CodeQL is free only for public code and this project is private, so I used Semgrep, which runs on my own machine.

Each kind of data lives in a different place. Roles are held in Keycloak; the application reads them from the token. Patient records come from ANZER, which this project reads but does not own, change, or assess. Case data and the audit trail sit in the application's own PostgreSQL database. The FHIR server is bound to the local interface, so no other machine can reach it.

5. Implementation

This chapter describes what was built: signing in and how sessions are handled, what each user may do and how that is enforced, how their actions are recorded, and the security of the software the system runs on.

5.1 Identity and Access Management

5.1.1 Sign-in and Session Handling

Users sign in through Keycloak, an identity provider running on the hospital's own servers, using OpenID Connect, a standard way for two systems to agree who someone is. The application never sees a password. It sends the user to Keycloak, and Keycloak sends back a token saying who they are and what role they hold. Password rules, account lockout, and the accounts themselves all live in Keycloak.

Roles are held in Keycloak and nowhere else. The application reads the role from the token and keeps no copy of its own. That is why there is no screen for editing permissions: to change what someone may do, an administrator changes their role in Keycloak. Five roles exist: Surgeon, Anaesthetist, Surgical Coordinator, Finance, and System Administrator; every token must match exactly one of them.

If a token carries no valid role, sign-in is refused. The user is not let in with fewer rights, they are not let in at all. This was not how it worked at first: such a user used to get a working session and could reach every page that asked only for a login. The check now runs in three places: when the user signs in, before any screen

appears, and on every request the server receives. The refusal itself is not written to the application's log, because sign-in stops before anything is recorded, so repeated attempts show up only in Keycloak's own log.

Figure 4: The sign-in path. Keycloak checks the password and returns a token carrying the user's identity and role. A token with one valid role gets a session; a token with none is refused.

Authentication (Keycloak): User signs in → Keycloak (passwords, logout) → **token** → Authorisation (the application): Resolve the role from the token (one of five roles, or none) → Session issued (one valid role, eight-hour token) / Refused (no valid role, no session created)

The application holds no passwords. Roles are held in Keycloak and nowhere else. A refusal is recorded only in Keycloak's own event log, because sign-in stops before the application writes one.

The server does not keep track of who is signed in. Instead, each user gets a token that proves who they are for eight hours, shortened from the library's default of thirty days. Because there is no server-side list, a token cannot be cancelled early; it stays valid until it expires. Keycloak also keeps its own separate session, timing out after twenty minutes of no activity or eight hours in total. The application adds a further twenty-minute idle timer of its own, with a one-minute warning. This one only runs in the browser: it is a visual timer, not something the server checks. It only catches someone who leaves the tab open and walks away. If the tab is closed instead, even briefly, the timer forgets that time passed and starts again from zero the next time the tab opens. So closing the tab does not stop a different person from opening the browser soon after and landing straight in. The timer also has no effect on the token's own eight-hour lifespan.

Two problems with this design have since been fixed. First, signing out now properly ends both sessions instead of just the application's. Before this fix, the next person on a shared computer could still be signed in as the last user, because only the application's side was cleared. Second, the two systems' timers were brought into line, so neither outlasts the other.

One limitation remains. Because there is no server-side session record, nobody can cancel a token early, and idle time cannot be enforced by the server. A disabled account is no longer part of this problem, since that is now checked directly on every request, described in the next section. But a genuinely stolen token would still work for the rest of its eight hours, since there is still no way to cancel it early.

5.1.2 Account Status and AD Federation

Keycloak originally held its own standalone accounts, created and managed only inside Keycloak itself, separate from any hospital system. During this work, Keycloak was connected to the hospital's own Active Directory through user federation instead (section 4.4 describes this decision). Accounts and their enabled or disabled status now come from AD, so a person's access in this system follows the same account AD already manages for them, rather than a separate copy Keycloak held on its own. This made it possible to check status live, on every request, instead of only at the occasional point a standalone account happened to be read.

Access now requires two conditions to hold at once: the account must be enabled in Keycloak, and it must be enabled in the application's own record. The check is centralised in a single function that every entry point calls, so the four surfaces that enforce it (sign-in, page navigation, and the authorisation helpers every API route uses) can never disagree with one another. The two conditions are not symmetric. A Keycloak-side disable overrides everything: if Keycloak reports the account disabled, the user is locked out regardless of what the application's own record says, and no administrator can override that from inside the application. An administrator can still enable or disable an account independently through the application's own screen, and that setting is enforced on its own terms whenever Keycloak's side says enabled: an application-side disable still blocks the user even though Keycloak allows them. So the rule is closer to a logical AND than an override: both sides must say enabled, but only one side, Keycloak, can force a lock-out the other side cannot undo.

A further condition covers users who disappear from Keycloak or AD entirely rather than being explicitly disabled: such an account is marked disabled in the application with a distinct "not found" status, and if that person's account later reappears, they are issued a new account rather than having the old one reactivated, a decision made jointly with the hospital's IT team since silently relinking an old account to a returning identity was judged a greater risk than the inconvenience of starting fresh.

Figure 5: Account status after AD federation. Access requires two conditions to hold at once, checked on

every request: the account must be enabled in Keycloak, whose status is itself now sourced from AD through federation, and enabled in the application's own record.

User enters AD credentials → Active Directory (validates credentials) → Keycloak (federated identity, issues token) → Application checks: Keycloak-side enabled? AND application-side enabled? → Both true: access granted / Either false: access refused, refusal written to the audit trail

A Keycloak-side (AD-sourced) disable always locks the user out, even if the application's own record says enabled. An application-side disable also blocks access even if Keycloak allows it. Only Keycloak can force a lock-out the application side cannot override.

This design replaced an earlier approach that did not work. Account status was originally meant to be enforced through a scheduled job: read each user's status from Keycloak on an interval, and disable anyone Keycloak reported as disabled. That job never ran, because its trigger was written for a hosting platform this deployment does not use, and this was recorded as an open gap.

The live check closes that gap, and does so more completely than the original job would have: a periodic job still leaves a window between checks where a disabled account remains usable, while a live check on every request has no such window. The check does not end an already-open session file the instant an account is disabled: the browser tab stays open, and a request already in flight at the exact moment the account is disabled still completes, which is an unavoidable limitation of any real-time, database-backed check rather than a gap in the design. But the user's very next request to the server, whether that is loading a new page or saving a form section they were in the middle of, is rejected before any write logic runs, with an "account disabled" response, and the refusal itself is written to the audit trail. Actions genuinely completed before that point remain correctly attributed: disabling an account part-way through a session does not alter what was already recorded. The other two background jobs described in section 5.4, unrelated to account status, were not part of this change; their own outcomes are described there.

The two places were not originally recorded equally. Changes made on the application's screen were written to the audit trail; changes made directly in Keycloak or AD were not, and an investigation into an account's status had to look in two places. This was since closed by configuring Keycloak's event system to forward selected event types into the application's own log: account enabled/disabled, and failed sign-in/sign-out attempts at the Keycloak level. Keycloak's event listener supports far more event types than these; the remaining ones were left as an option for the hospital's IT team to enable later if they judge them useful, rather than added speculatively now.

One finding here mattered more than it first looked. The session carries the identifier Keycloak issues, while the application's own database uses its own row identifiers for the same people. Checks of whether a case belonged to a particular doctor compared the two directly, and two different kinds of identifiers never match, so doctors could not open their own cases. It appeared as several unrelated faults before the single cause was found: a list showing other people's drafts, a saved draft opening read-only, and errors on writing. This was corrected, and the checks now resolve one identifier to the other before comparing.

5.2 Access Control Model

Access control in this system combines the two models described in Chapter 3. Roles carry the decisions that follow a person's job, attributes carry the decisions that depend on a particular case, and the case's position in the workflow carries the decisions that depend on how far the work has progressed.

Mapped onto the four inputs NIST describes [3], the model uses three of them. Subject attributes are the user's role, taken from the Keycloak token, and whether their account is active. Object attributes are the identifiers of the doctors assigned to a case and the step that case has reached. The rules are the forty-two action rules described below. Environment conditions are not used: no decision in this system depends on the time of day, the location of the request, or the network it arrives from.

Table 4: The five roles and what each is responsible for

Role	Responsible for
Surgeon	Creating a case, entering the clinical description and their own fee

Anaesthetist	Taking an unassigned case from the pool and adding their assessment
Surgical Coordinator	Monitoring progress, and setting case priority
Finance	Building and confirming the cost, submitting the case, recording patient consent, finalising it, reviewing and sending the request to the insurer, and recording the insurer's answer
System Administrator	Managing user accounts, reading the audit trail, viewing all cases and patient data, assigning doctors to cases, and using emergency override powers (releasing a stuck case, removing documents)

Forty-two actions are defined across the system, and each has a rule saying which roles may perform it and under what conditions. They fall into four groups: eleven for the case lifecycle, six for finance and the insurer, six for documents and notes, and nineteen for reports, administration and reference data. The rules are kept in a separate document, updated as the code changes. It reached its third version during this work, each version following a round of checking.

The three factors work together, not separately. All three must hold, and if any one fails the action is refused. Take one question: may a surgeon edit this case? Using the role alone, the answer is yes, but that would let any surgeon edit any case. Adding assignment narrows it: yes, if the case is theirs. Adding the step narrows it again: yes, if the case is theirs and it has not yet reached Finance. Only the third answer is correct. The step is the least obvious of the three, because it changes the answer for a user who has done nothing: when the anaesthetist finishes their part, the case moves to Finance, and the surgeon becomes read-only. Figure 6 shows the three factors as they are evaluated for one request.

Figure 6: The three factors evaluated in every access decision. All three must hold; if any one fails, the action is refused.

Role: Surgeon (role-based access control) → Assigned to this case (attribute-based access control) → Case has not reached Finance (the case's step in the workflow) → → **Access decision** (checked in one module) → **May edit the case** (refused if any factor fails)

All three must hold. If any one fails, the action is refused. The third can change the answer even when the role and the assignment have not.

Underneath these rules is a split the hospital already used on paper. The doctors say what treatment is needed and what their own fee is. Finance builds the final price from that, adding or removing items as the case requires, and commits the hospital to the total. Neither does the other's job: a doctor cannot set what the hospital charges, and Finance does not decide what treatment is given. The case's step is what keeps them apart in software. Without it, a doctor could change a figure after Finance had already settled the price.

The Surgical Coordinator monitors progress and sets case priority; no role holds an edit right over another role's part of a case. The same separation applies to what leaves the hospital; Finance reviews a case before the request is sent, so nothing reaches an insurer without a person having checked it.

Assignment itself needed a rule of its own for the Anaesthetist role. Early on, opening an unassigned case made it the anaesthetist's immediately, with no typing and no saving required, and there was no way to hand it back if it had been opened by mistake. A case is now taken only when the anaesthetist saves or verifies it, and they can release it while it is still a draft. This matters for the same reason the case's step matters elsewhere: an action with a real consequence, becoming responsible for a case, was previously happening as a side effect of merely looking at something.

Access is decided in one place. A single module checks that the user has a session, that their account is active, and that their role is valid. Where the action needs it, the same module also checks that the case is theirs and has reached the right step. Nearly every address in the system calls it. Not every part uses the full check, though: at least eight places ask only whether the role is right, then work out for themselves whether the case belongs to the user, using their own separate copy of that rule rather than the shared one. This has already caused a real problem once, not just a theoretical risk: one of those copies fell out of step with the shared rule and briefly let a surgeon reach a case they should not have been able to see, before it was found

and corrected. The other copies currently still agree with the shared rule, kept in sync only by developers remembering to update every copy by hand. Chapter 6 returns to this.

5.3 Audit Logging

The audit log records every action that changes a case. An entry holds who acted, the role they held at the time, which case was affected, what was done, when, and which part of the system was used. Refused attempts are recorded too, with the reason. The entry does not record the device or the network address, so it can show that an account acted but not which machine was used.

The record is append-only: entries can be added, but never changed or deleted. This is enforced at the database level, through a trigger that rejects any update, delete, or truncate against the table outright, combined with a permission grant that removes the application's own database login's ability to attempt those operations at all. Neither the System Administrator role inside the application, nor a compromised application process using its normal database credentials, can alter a log entry; both are blocked before the operation would even reach the trigger. OWASP recommends exactly this for records that need to be trusted [6]. It is what makes the log usable as evidence, rather than just something to look things up in.

One honest caveat sits underneath this, distinct from anything reachable through the application: whoever holds the database's own superuser credentials (a different, infrastructure-level actor entirely from any role the application defines) can disable the trigger with an explicit administrative command before modifying rows, then re-enable it. This is not a gap in the design so much as an inherent property of database superuser access generally, and it requires credentials no application role has access to. It happened exactly once, in a local development database, as an authorised, one-off cleanup of test data, carried out as a deliberate and distinct action rather than an ordinary delete, and never applied to the deployed system. Those credentials were handed over to the hospital at the end of the placement; the development team retains no database access on the live system. In short, the immutability guarantee described above is an application-level guarantee, holding against every role the application defines; who controls the database's own infrastructure-level superuser credentials is a separate governance question entirely, answered by who holds server access, not by anything this application can enforce in code.

The role is recorded at the moment of the action rather than looked up later. This was corrected during the work: Finance and the Coordinator had both been logged as STAFF, so the record could not show which of them had acted. Stamping the exact role at the time of writing also means that a later change of role does not alter what past entries say.

The administration screen presents a second log alongside it, and the two are not the same thing. The audit log records workflow events, as described above. The activity log records every successful request that passes the access checks, including the request body, so before a fix described below, it accumulated patient identifiers and clinical detail as a side effect of recording traffic. It originally had none of the protection the audit log has either: its rows could be changed or deleted, and no retention period was set, and this was recorded as an open gap. It has since been given the same append-only design as the audit log, and is in fact slightly stricter: the audit log's trigger allows one narrow, deliberate exception, where deleting a case correctly nulls out that entry's case reference rather than leaving a dangling link; the activity log has no foreign-key relationships of this kind, so its trigger rejects every update, delete, and truncate outright, with no exception at all. Separately, the patient and clinical fields the request body carried are now redacted before the row is written, replaced with a fixed marker rather than stored in the clear; a small number of fields, such as city, country, and gender, are not yet on that redaction list and would still be stored in full if a request happened to carry them, which is recorded as an open item in section 7.3.

Two limits remain. Neither is a fault in the sense of a broken control; the second is a genuine, unresolved inconsistency worth stating precisely. The first is names, and it is more mixed than a single description covers. The audit entry's actor field is a reference to the user, resolved to a display name only when the record is read, so a later change to that person's name changes how every past entry involving them is shown. Separately, many entries also embed a name as literal text inside their own detail, "Case finalised by John Tan (SURGEON)," for instance, captured at the moment the action happened, and this text is genuinely frozen: it is written once into the append-only log and the database itself rejects any later change to it. The result is that a single entry can show two different names for the same person if they renamed in between: the frozen name in the detail text, and the current name in the actor field next to it, disagreeing with each other on the same row. Users can no longer rename themselves through the application, though names synced in from

Active Directory can still change this way, so the inconsistency remains reachable. The same current-name behaviour reaches insurer emails, printed forms and the coordinator export, so a rename can also affect documents that already left the hospital, independent of the log itself. The second limit is who may read the log. Administrators open it through the administration screen built as part of this work, and their own actions appear in both logs, so they can see what has been recorded about them. Separating those two responsibilities would mean redesigning who can read what, not fixing a fault.

Four things are not written down. A refusal that means only "not yet" (submitting a case before it is ready, or deleting one that has moved past draft) comes from the workflow rather than from a permission check, and only permission refusals are written down. A request for a case that does not exist is refused silently. A refused sign-in is recorded nowhere in the application, because sign-in stops before anything is written, so those attempts exist only in Keycloak's log. And nothing at all can be written when the database itself is down.

Nothing checks the log as it fills, and nobody reviews it on a schedule. Refused attempts are written down correctly, but no one is told, so twenty attempts in a row would sit in the table until somebody happened to open it. OWASP renamed this category in 2025 to include alerting for that reason [13]. Both are recorded in section 7.3: the software that would raise an alert, and the hospital routine that would review the records.

5.4 Deployment

My teammate deployed the application, and the hospital's IT department provided the server. What follows is about the security of that environment: what I found in its configuration, what was corrected, and what is still open.

The system runs on a Windows server the hospital owns, reachable only from inside the hospital. Four programs run on it side by side: the application, the database, Keycloak, and the FHIR server. None can be reached from the internet, and the FHIR server can only be reached from the machine itself. Patient data is read from the hospital's own system, with limited read-only access granted by the IT department, so this deployment does not own or hold the primary patient record; ANZER remains the single source of truth for that data, as section 4.4 states. This does not mean patient information never appears inside this system's own storage. As section 5.3 describes, the activity log records the body of every successful request, and some of those requests carry patient identifiers and clinical detail; that information was originally copied into the activity log as a byproduct of logging traffic, and is now redacted before the row is written, with a small number of fields not yet covered by that redaction, recorded in section 7.3. I did not test whether the system keeps running under normal use, and that is recorded in section 7.3.

The application has a shortcut for developers: skip sign-in, pick a role, no password. It is useful on a laptop and dangerous on a server. The template that people copy when setting up a new server had that shortcut switched on, so anyone following it would have built a server that anyone could walk into. This was fixed twice. The template now has it switched off, and the application refuses to start if the shortcut is on in production.

The browser-facing connection now runs over HTTPS. Two internal connections remain unencrypted: Keycloak to the application, and Keycloak to Active Directory, so credentials and status information passing over those two links still cross the network in the clear. Keycloak also cannot leave its development mode until its own connections are encrypted, and nobody has checked whether the database on disk is encrypted. Turning on the remaining encryption is the server administrator's job, not mine.

The four containers are third-party software, so rather than assume they were safe I scanned them, and Chapter 6 reports what that found. One problem was found without a scanner. Keycloak had not been told where to keep its accounts, so it had defaulted to H2, the lightweight, file-based database built into Keycloak itself, meant only for local testing and explicitly not recommended by its makers for a real deployment. H2 stores its data as a single file inside the Keycloak container rather than as a separate, durable service, so losing that container would have meant losing every account outright, and the file could not be backed up without shutting Keycloak down first to release it. Keycloak was reconfigured to use the PostgreSQL database already running alongside it instead, in a separate schema with its own restricted login, so accounts now live in the same durable, backed-up database as the rest of the system rather than a throwaway file tied to one container's lifetime.

Three background jobs existed at the time this was written: the account-disable job described in section 5.1.2,

one to warn when a case is taking too long, and one to close cases automatically once past a deadline. None of them ran, for the same reason in every case: all three declared their schedule in a file meant for a hosting service this project does not use, and nothing on the hospital's server started them. One was also called from the browser when the application was open, but that call used a method the route did not accept and was rejected before any of it executed.

Each of the three has since taken a different path. The account-disable job was superseded rather than fixed, by the live dual-check design described in section 5.1.2. The deadline-closure job was retired outright as a feature rather than scheduled, it closed cases after a flat 30-day window unrelated to the SLA tiers described below, and it was deliberately removed from the codebase; a database migration later confirmed it had auto-closed zero cases across the system's history before being deleted. The slow-case warning job was kept and given a working trigger, but not the one originally planned: rather than an OS-level scheduler such as Windows Task Scheduler, it now runs on an in-process timer inside the application itself, firing automatically every 15 minutes for as long as the application process is running. This closes the "never triggers" problem for as long as the process stays up, though it is a different, less independent guarantee than an OS-level schedule would give: if the application process stops, the timer stops with it, whereas a Task Scheduler entry would not depend on the application process being alive at all.

The rule the slow-case warning job checks against was also revised after this was first written. Cases originally fell into two service-level categories, adjustable from a settings screen. This was changed to three fixed categories: Emergency (1 hour), Urgent (8 hours), and Routine (48 hours), read from a database-backed configuration rather than the earlier hardcoded pair. The permission to change these three thresholds was built for the Surgical Coordinator and System Administrator roles, with validation, a database write, and its own audit entry, and this was tested working.

It is not usable today: the screen that edits it and the route behind it are both switched off behind a single flag, deliberately, after two real bugs were found in the editor during testing, a saved total that was written but never read anywhere, and a dashboard countdown computed from a separate table the editor did not update. Rather than ship a partially working editor, both the screen and the route were locked pending a decision, and the route refuses the request outright regardless of who is asking, so a disabled feature can never be mistaken for a wrong-role refusal in the logs. Reading the current thresholds remains open to everyone; only changing them is switched off. So the thresholds are fixed for every role today in practice, though not by design: the capability to adjust them exists, built and tested, waiting on that flag.

The job raises its warning in stages against these thresholds rather than only at the deadline, and the warning itself is system-level only: it surfaces as a status or flag inside the application rather than being sent out as an email. These tiers apply only to this warning job; the separate deadline-closure feature that once existed worked on an unrelated flat time window and, as noted above, was removed rather than connected to this newer tier system.

6. Results and Discussion

This chapter reports what the security work produced, and what those results mean. Section 6.1 gives the findings from both checking methods, the numbers each produced, and the decisions taken on them. Section 6.2 looks more closely at two of those results: how permissions in this system depend on more than the user's role, and why the automated tools did not find the problems that manual testing found.

6.1 Results

6.1.1 How the System Was Checked

The system was checked in two ways. Automated tools looked at three things: the outside packages the system uses, the container images it runs in, and the project's own code. Manual testing was done role by role on the running system, using one account for each role and following each case from creation to the insurer's decision. Both were done in the same period, on the same version of the system, and the rules being tested covered forty-two actions across five roles. The ASVS assessment described in section 6.1.6 was carried out separately, later in the work, against a subsequent version of the system.

Two checks were still to come at the time of this assessment. The hospital's IT Security Division had not yet

tried to break into the system, and the container settings had not yet been checked against the CIS Docker Benchmark. Neither had been done when this was written. Both have since occurred: Canada Bank (IT security department) carried out a penetration test jointly with the hospital's IT Security Division, and separately reviewed the project's code and its containers. Their findings are confidential and are not reproduced in this thesis.

6.1.2 Package and Container Scanning

Table 5: Scanning targets and findings, versions as scanned at the time

Target	Tool	Findings
npm packages	npm audit	15: 2 critical, 4 high, 8 moderate, 1 low
Keycloak 26.6.4	Trivy	16 high, 0 critical
PostgreSQL 16	Trivy	72, then 55 after an update
HAPI FHIR 7.4.0	Trivy	132: 27 critical, 105 high

The versions shown above are the versions in place when each scan was run, not the versions in place today; Table 3 in section 4.4 gives the current version of each tool. Keycloak has since been updated again, from 26.6.4 at the time of this scan to 26.7.2 currently.

Every problem was in software the project uses rather than in code the team wrote: third-party libraries and container images. Updating the login library cleared both critical findings, taking the count from 15 to 13 with none critical. Eight of the remaining problems were in the application framework, and looked at first as though they needed a major upgrade. They did not: moving from version 15.5.20, the version in place before this fix, to 15.5.22 was enough to clear them at the time. Next.js has since been updated again as part of routine maintenance, unrelated to this specific finding, and is at 15.5.24 as reported in Table 3. One package, sharp, is still below the version its advisory asks for, because the fix offered would have undone the framework update. That was judged a step backwards, and it is recorded as an open item.

Updating the database image reduced its findings from 72 to 55, and its critical findings from 19 to 17; the rest have no fix available yet, so nothing can be installed. The FHIR server image was the worst of the three, and whether it is needed at all was raised as an open question for the team rather than treated as something to patch. Keycloak was also found to be keeping every account in its own built-in database, which its makers say not to use for real systems. It was reconfigured to use the PostgreSQL database already running alongside it, in a separate area with its own restricted login. The accounts on my own machine moved across keeping their original identifiers; any other installation starts with an empty Keycloak and needs its accounts created again.

6.1.3 Checking the Project's Own Code

Semgrep was used for this step; section 4.4 explains the choice over CodeQL. It applied 257 rules across 356 files and returned 15 findings. secretlint was also run over the repository and its files, checking for secrets left in the code such as passwords or API keys; it found none.

Table 6: What Semgrep found

Type	Count
Path traversal, looked at separately	1
Regular expression built at run time	3
Unsafe text in log messages	5
Unpinned action versions	4
Installer piped into a shell	1

Package setting	1
-----------------	---

None of the common problems appeared. There was no SQL injection, no cross-site scripting, no password left in the code, and no way around the login. Nothing was about access control.

The path traversal was looked at more closely and could not be used. The database check runs before any file is written, and an invalid case identifier fails there, so the write is never reached. That protection was accidental rather than planned, so it was recorded as something to make deliberate: check the identifier properly, and mark the order of steps as security-critical so a later change does not remove it by accident.

6.1.4 Manual Role-by-Role Testing

6.1.4.1 What Was Found

A finding was counted as access control if it changed which users could do which action in which case. Findings about audit logging, sessions, input checking, data quality, and screen behaviour were counted separately. The logout defect was counted as access control rather than as a session problem, because its effect was that one user reached another user's account without logging in. On this basis the security phase recorded 47 findings, of which 23 were about access control.

Table 7: The 47 findings by type

Type	Count	Example
Access control	23	See Table 8 for the six sub-categories
Screens, data quality, workflow	16	Field validation, labelling, and workflow-order issues not related to permissions
Audit and logging	4	e.g. refused actions not written to the log
Input checking	1	A single validation-related finding; specific detail not retained
Sessions	1	Logging out left the Keycloak session open, letting the next user at a shared computer in without a password (fixed)
Deployment and configuration	1	Keycloak had defaulted to its own built-in H2 database instead of the PostgreSQL database alongside it (fixed)
Closed, not a defect	1	A finding investigated and confirmed as intended behaviour, not a bug; which one is not retained

The 23 access-control findings fell into six groups.

Table 8: The six kinds of access-control finding

Group	Example
Seeing cases they should not	A surgeon's list showed other surgeons' drafts
Actions held by the wrong role	Submit and finalise sat with the Coordinator, and refused Finance, the role that owns them
No lock on the case's step	Doctors could still edit details and costs after Finance had reviewed the case
Wrong identifier compared	Assignment checks compared two different kinds of user id, so doctors could not reach their own cases
Letting people through on failure	The disabled-account check allowed access when it found no record

Sessions outliving logout	Logging out left the Keycloak session open, so the next person at a shared computer entered the previous user's account
---------------------------	---

6.1.4.2 What Was Done About It

Most findings were fixed before deployment. Twenty-three were recorded as decisions to leave for later. A representative sample is given below, each with the reason it was deferred; a full individual write-up was not retained for every one of the twenty-three.

Table 9: Examples of work left for later, and why

Left for later	Status	Reason
Duplicate drafts	Deferred	Detection is built and working: a surgeon starting a case is warned of an existing draft and can jump to it. Deleting the old draft is possible but through a separate, general delete action, not linked to that warning; merging two drafts into one was never built
Reversal (formerly "appeal")	Deferred	Built as a complete feature, allowing Finance or an administrator to reopen a closed case and re-enter the workflow without starting over, but switched off behind a single flag pending a decision to enable it
Refused sign-ins not recorded in the application log	Deferred	Sign-in stops before the audit write; the attempts exist in Keycloak's log
Audit log's recorded address can be set by the client	Deferred	The reverse proxy appends the real client address to an existing header rather than replacing it, and the application reads only the first value in that header, which a client can set to anything before it reaches the server
No database-level rule stops a surgeon's id from being written into the anaesthetist's slot	Accepted	Every route that assigns a doctor already blocks this by role, so it is not reachable through the running system today; the rule exists only in application code, not in the database schema, and adding a schema-level constraint was judged low priority given no live exploit path
A clinical-form edit after the surgeon has verified can leave no audit trail	Deferred	A form-only save is deliberately not logged while a case is still in draft, so a save made in the window after verification but before the case reaches Finance goes unrecorded

The status shown above reflects a judgement made about each of these six specific examples, applied when this table was assembled. It was not recorded item by item at the time for the full twenty-three, so this classification does not extend to the seventeen items not shown here.

The clearest illustration of a finding that carried more weight than it first appeared to have concerned the page-level access rules. The file holding those rules sat at the top of the repository, while the application runs from a folder inside it. The framework only looks inside that folder, so it never found the file, and those rules had never run at all. No additional exposure was demonstrated from the inactive page guard, because server-side route checks described in section 5.2 remained in place; however, this depended on those lower-layer checks being correctly and consistently implemented, not on any property of the missing guard itself. What only emerged later was that the file's own settings excluded those addresses entirely, so even had it loaded, it would never have guarded what it appeared to guard. It read as a control, and was never capable of being one. It was left as it stood: switching on a gate that has never run means every page might behave differently, and that needs a full test pass across the whole system.

A related finding was fixed at once. A user who belonged to no group still received a working session, and could reach any page and any address that only asked whether someone was signed in, including a case list that returned every case. Sign-in is now refused when no valid role is found.

Two findings concerned the point at which work should stop, and both were fixed. Doctors could change the diagnosis and the cost after Finance had reviewed the case, without Finance knowing, so Finance had committed to a figure the case no longer matched. Editing now stops the moment a case reaches Finance. This is the finding that shows the case's step has to be part of the rule: the doctor's role has not changed, and the case is still theirs, but the answer must change anyway. Separately, opening an unassigned case made it the anaesthetist's immediately, with no typing and no saving, and there was no way to hand it back; this is described in section 5.2, where the take/release rule that fixed it is set out alongside the rest of the access control model. That release mechanism produced its own smaller bug later: a case released by one anaesthetist could be silently reclaimed a few seconds afterwards by a leftover autosave request still in flight, quietly bringing back the data the release was meant to clear. The fix checks for a recent release before allowing a reclaim, and fails closed rather than open if that check itself cannot run: a database error blocks the reclaim rather than letting it through.

Logging out cleared only the application's own session while the Keycloak session stayed open, so the next person at a shared computer entered the previous user's account with no password. Signing out now ends both.

The log had a gap too: it was written only after a successful action, so a refused attempt left nothing behind. This was partly closed. Refusals where someone reaches for a case that is not theirs are now recorded, with the person, their role, the case and the reason. Refusals that only mean "not yet" are still not recorded, and neither is a deletion refused because the case has moved past draft.

A further gap sits earlier. When someone with a real account but no role is refused at sign-in, nothing is written to the application's log at all, no success, no failure, because sign-in stops before the point where anything is recorded. Those attempts appear only in Keycloak's own log, which is switched on, so an investigation has to look in two places rather than one.

Two further weaknesses concern the audit trail itself. Records hold a reference to the user rather than the name as it stood at the time, and look the name up when the record is read, so a rename relabels every past entry for that person, including sign-ins and refused attempts. A user renaming themselves through their own profile was closed, but an administrator creating or renaming an account, and a later sync that pulls a person's display name from Keycloak on each sign-in, both still trigger the same relabelling, so the underlying gap remains open despite that one path being closed. Separately, administrators can open the audit trail while their own actions are recorded in it, so a privileged user can see what was captured about them. Both are design changes rather than fixes, and both are recorded for later. Neither would be found by a scanner, and role-by-role testing did not flag them either, because the system behaves exactly as it was built to.

Two findings concerned patient data. Neither is fully resolved, though both have moved since. Removing a patient identifier from the screen did not remove it from the data sent to the browser. Email, city, and country have since been dropped from that response, but the national ID and phone number are still sent even when the screen does not display them. Fully narrowing this to only what each screen needs, and moving the search itself to the server, was judged too large a change to complete before go-live, and remains a separate, careful pass still to be done.

The second concerned transport. At the time, the system ran entirely over plain HTTP, which the browser reported directly: it showed the connection as not secure. The browser-facing connection has since been put behind HTTPS. Two internal connections remain unencrypted: Keycloak to the application, and Keycloak to Active Directory. Patient data and credentials passing over those two links still cross the hospital network in the clear. This is a partial departure from a standard the project otherwise follows, since FHIR's security guidance states that production data should be sent over an encrypted connection [2]. Encrypting the remaining two connections is a server configuration rather than an application change, and it is recorded in section 7.3.

Two further findings came to light during the final review, after the counts above were settled. The first concerns a second log, already introduced in section 5.3: the activity log accumulated patient identifiers and clinical detail as a side effect of recording traffic, and at the time it had no protection against being changed or deleted, unlike the audit trail. Both gaps have since been closed: it has been given the same append-only protection as the audit trail, and the patient and clinical fields it carries are now redacted before being written, with a small number of fields not yet on that list, recorded as open in section 7.3. The second concerns the background jobs described in section 5.4: three existed, and at the time none had ever run. Each

has since taken a different path. The account-disable job was superseded by the live dual-check design described in section 5.1.2. The slow-case warning job was kept and given a working trigger, an in-process timer rather than the originally planned OS-level scheduler. The deadline-closure job was retired outright rather than scheduled.

6.1.5 OWASP Top Ten Assessment

Every category of the OWASP Top 10:2021 was rated against evidence from this project. Categories were rated strong where the risk was assessed and no significant gap remains, partial where controls exist but a known gap is recorded, and needs work where a material weakness is unresolved.

Table 10: OWASP Top 10:2021 ratings

Rating	Categories
Strong	A07 authentication
Partial	A01 access control, A02 cryptography, A03 injection, A04 design, A05 misconfiguration, A06 outdated components, A08 data integrity, A09 logging, A10 request forgery
Needs work	-

No category was rated as needing work, though several of the partial ratings hold real, current gaps. The major access-control defects identified under A01 were substantially remediated through role-by-role testing rather than by a scanner: object-level scoping and the case-assignment checks were fixed, and the Coordinator's access was narrowed to its own scope. Remaining open items prevent the category from being rated Strong. System Administrator currently has write access to clinical form data and cost figures through two routes, despite the system's own design stating this role should be view-only there; this is tracked as an open item in section 7.3.

A02 covers encryption in transit and at rest: the browser-facing connection now runs over HTTPS, but two internal connections, Keycloak to the application and Keycloak to Active Directory, remain unencrypted, and nobody checks whether the disk is encrypted. A05 covers configuration hardening. Real security headers are set, and required secrets fail loudly rather than falling back to an insecure default, but Keycloak runs in its development mode even in the production deployment path, which its own documentation advises against, and no automated dependency or code scanning runs in the build pipeline to catch drift between releases. A fourth A05 item was fixed during the phase: Keycloak's built-in database, which was replaced by the PostgreSQL database already running alongside it.

A09 rests on an audit trail that cannot be changed or deleted, protected by the database itself, and that records refusals along with the reason. At the time of this assessment, two gaps sat against that: nothing watched the log as it filled, and the activity log held request bodies without the same protection as the audit trail. The activity log has since been given the same append-only protection, closing that specific gap. What remains is narrower but still real: a sign-in refused for having no valid role writes nothing to either log, only to a server console outside the application's own record, and nothing watches either log as it fills.

A06 rests on scanning and fixing packages and container images, and was strong immediately after that work. It did not stay that way: a later dependency check found new advisories that were not present right after the fix, because no automated scanning step runs in the build pipeline to catch this drift between releases. The components in active use are still current; what is missing is a repeatable check, not a one-time fix, and that gap is recorded in section 7.3. A07 relies on Keycloak, the password rules, account lockout and session handling, now strengthened further by completed AD federation and a live per-request account-status check; there is no multi-factor authentication, which was deliberately left to a later phase, but nothing found weakens this rating.

6.1.6 ASVS 4.0.3 Assessment

Alongside the OWASP Top 10 review, the system was separately assessed against the OWASP Application Security Verification Standard, version 4.0.3, targeting Level 2 [14]. ASVS is a more granular standard than the Top 10: rather than ten broad risk categories, it sets out roughly 280 individual requirements across

fourteen chapters, covering areas the Top 10 does not break out on its own, such as session management, error handling, and business logic.

ASVS 4.0.3 was the formal baseline selected for this project and applied consistently throughout, at Level 2, across all fourteen chapters. A newer version, ASVS 5.0.0, became available during the project, released in May 2025, but the assessment already under way continued against 4.0.3 rather than switching partway through. Re-assessing the system against ASVS 5.0 is recorded as future work in section 7.3.

All fourteen chapters were assessed; none were skipped wholesale, though a small number of individual requirements were marked not applicable where the underlying feature does not exist in this system, for example, the requirements covering SOAP and GraphQL interfaces, since neither is used here, and self-service credential recovery, since accounts are provisioned rather than self-registered. Sixty-two subsections were checked in total: 29 were met, 20 partially met, 6 not met, and 7 not applicable. Level 2 was not fully achieved; no chapter was rated as fully met.

Table 11: ASVS 4.0.3 results by chapter

Chapter	Met	Partial	Not Met	N/A	Subsections
V1 Architecture, Design and Threat Modelling	5	3	1	0	9
V2 Authentication	2	3	0	2	7
V3 Session Management	5	1	0	1	7
V4 Access Control	3	0	0	0	3
V5 Validation, Sanitization and Encoding	2	2	0	1	5
V6 Cryptographic Storage (at Rest)	1	1	1	1	4
V7 Error Handling and Logging	0	3	1	0	4
V8 Data Protection	1	1	1	0	3
V9 Communications	1	1	0	0	2
V10 Malicious Code	2	1	0	0	3
V11 Business Logic	0	1	0	0	1
V12 Files and Resources	5	0	1	0	6
V13 API and Web Service	0	2	0	2	4
V14 Configuration	2	1	1	0	4
Total	29	20	6	7	62

Each of the six not-met subsections sits in a different chapter, with no chapter failing more than one: no documented threat-modelling process or automated security scanning in the build pipeline (V1); patient data held in plain, unclassified columns with no field-level encryption (V6); no centralised log monitoring or alerting (V7); no data-retention or deletion policy (V8, the same gap noted below as deliberately deferred); no malware scanning or integrity check on uploaded files (V12); and minor server-fingerprinting headers left at their defaults (V14). Taken together, the weakest areas cluster around process and lifecycle gaps rather than exploitable code-level defects: nothing here is a working attack, but each is a control the standard expects that this project does not yet have.

Fifteen findings were tracked (F1 through F15). Ten were resolved or substantially addressed during this work, including a critical Keycloak vulnerability allowing account takeover through a flaw in its credential-reset flow, fixed by upgrading the Keycloak version; eight high-severity dependency vulnerabilities, patched to bring the dependency scan to zero high or critical findings at the time; patient identifiers that were being sent as URL query parameters rather than in the request body, moved across four call sites; and an access-control gap on the patient-lookup endpoint that had no case-assignment check at all, closed by routing it

through the same shared visibility check used elsewhere in the system. That dependency result did not stay at zero: a later check found new high-severity advisories in build-time tooling, disclosed after the original fix rather than caused by any reversal of it. No automated scanning step runs in the build pipeline to catch this kind of drift between releases, so a point-in-time fix does not stay verified on its own; this is recorded in section 7.3.

Three findings were left open at the time of the assessment. One concerns error messages that can leak internal detail to the browser rather than being suppressed outside of development. One concerned a developer sign-in shortcut that bypassed the normal login entirely. One concerns the lack of a step-up re-authentication requirement before high-consequence actions such as finalising or sending a request to the insurer. The remaining two are carried into section 7.3.

The developer sign-in shortcut was narrowed rather than fully removed. A loophole that let it activate outside a developer's own machine, on a staging server despite a production environment setting, was closed. The underlying shortcut itself is still present in the codebase, gated behind a flag that defaults to off and only takes effect in a non-production environment, such as a developer running the system locally. It is not reachable in the deployed system as configured, but it is not gone, and a future configuration change that enabled the flag outside a developer's own machine would bring it back. This is recorded as still open in section 7.3, not as resolved.

Two further findings were deliberately deferred rather than treated as defects. The first is the dead page-rule file already discussed in section 6.1.4.2: the unused file itself was removed, but building the real route-level enforcement layer it was meant to provide was set aside, since enabling nineteen untested rules across five roles risked silently locking a working role out of something it needs, without a full test pass to catch that first. The second is the absence of any data-retention or deletion policy, recorded here as what it is, an organisational decision for the hospital to make, not something a fix to the code could resolve on its own.

The assessment was carried out with AI assistance (Claude Sonnet 5), under my direction, consistent with the approach described in section 4.1: what to check, whether a finding was genuine, and what to fix or defer were my decisions, verified against the running system rather than accepted as reported.

6.1.7 How the Finding Counts Relate

Several counts appear across this chapter: forty-seven manual-testing findings, twenty-three of them access-control by type, twenty-three left for later by outcome, and twenty-three items in the Future Work table, and it is worth being explicit that they are not four steps in one shrinking pipeline. They come from slicing the same testing phase in different ways, plus one list from a different source entirely.

Manual testing produced forty-seven findings in total. Twenty-three of these are access-control findings, grouped that way by what kind of problem each one was, whether it changed which role could do what. Separately, twenty-three of the forty-seven were left for later rather than fixed before deployment, grouped that way by what happened to each one, fixed or deferred. These are two different ways of cutting the same set of forty-seven findings, one by type and one by outcome, and it is a coincidence of count, not the same twenty-three items, that both totals land on twenty-three. The other twenty-four findings were fixed before deployment.

The twenty-three items in the Future Work table, section 7.3, are a separate list again, not a subset of the twenty-three left for later. It draws from several sources: a curated selection of those twenty-three deferred manual-testing findings, folded together where more than one pointed at the same underlying gap; the three ASVS findings left open (error messages, step-up re-authentication, and the developer sign-in shortcut); and several items that were never part of the original forty-seven at all, raised later by the OWASP Top 10 re-check, described in section 6.1.5, and by design review, such as encrypting the remaining internal connections, moving Keycloak out of development mode, adding automated scanning to the build pipeline, extending the activity log's redaction to a few remaining fields, and consolidating the case-assignment rule's separate copies.

Table 12: Summary of manual-testing findings

Status	Count
Found	47

Fixed before deployment	24
Left for later	23

The twenty-three left for later are one combined category rather than two. Some are risks the project has accepted as they stand; others are real fixes simply not yet built. That distinction was not consistently recorded item by item at the time, so this thesis reports the single "left for later" status the working record actually supports, illustrated by the representative examples given above, rather than a finer split the underlying record cannot back up.

6.2 Discussion

6.2.1 Permission Depends on More Than the Role

Role-based access control asks one question: what is this user's role? In this system that was never enough on its own. Every rule rested on three things together, and removing any one of them gave the wrong answer.

Table 13: The three factors, with an example of each

Factor	Example from this system
Role	Only a surgeon may create a case
Assignment	A surgeon sees only their own cases; a doctor may remove only files they uploaded
The case's step	Once a case reaches Finance, both doctors become read-only

The third factor is the easiest to miss, because it changes the answer for a user whose role and assignment have not changed at all. A surgeon may edit their own case one day and not the next, not because anything about the surgeon changed, but because the anaesthetist has since finished their part and the case has moved to Finance. Neither the surgeon nor Finance did anything. A third person completing their work changed what the surgeon was allowed to do.

This was found as a real defect: doctors could change the diagnosis and the cost after the case had reached Finance, and Finance did not know.

That matters more than a single wrong permission, because it undoes a rule the hospital depends on. The doctors say what the treatment costs and Finance authorises it, and neither does both. If a doctor can change the figure after Finance has approved it, the separation exists on paper only. The case's step is what keeps the two apart.

The lock has a cost, and it was accepted knowingly. Once both doctors have verified, they are read-only. So if a correction is needed later, if the insurer asks for more clinical detail, for instance the doctor cannot supply it directly, and the change has to go through Finance or the coordinator. Uploading a document stops at the same point. The team judged this acceptable for a first phase, because the alternative is letting clinical data change underneath a price Finance has already confirmed. Whether to loosen the lock, and how, was left until after the go-live test.

Permission in this system depended on three things together, and the case's step was the one most easily overlooked. Most of the twenty-three access-control findings turned on assignment or on the case's step, not on the role. None of these three factors could have been known from a security framework alone. The case's step matters because Finance and the Coordinator agreed, over the weekly cycle described in section 4.1, that the doctors' authority over a case should end once Finance takes over, not because any standard requires a workflow-state check. The correctness of these rules rested on understanding the real work, collaboratively, before it could be translated into a check the code could enforce.

6.2.2 Why the Scanners Found No Access-Control Problems

Semgrep applied 257 rules to 356 files and returned 15 findings, none of them about access control. This is not a failure of the tool.

A rule in a scanner describes the shape of code. It can spot a query built by joining text together, or a password written into a file, because those look a particular way wherever they appear. "A coordinator may not submit a case" has no shape. It is a decision the hospital made about who is responsible for what, and nothing in the code marks it out from any other function call. A scanner cannot check a rule it was never given.

What this project adds is a measurement from a working clinical system. On the same code, in the same period, a general-purpose scanner found nothing in this category while role-by-role testing found twenty-three problems, several of them serious. Zhu and colleagues report the closest comparison: automatic analysis of six applications found a set of vulnerabilities, and when developers gave the tool the application's own access control rules, twenty more appeared that the automatic methods had missed [10].

A related gap appeared before any scanning was done. The access control rules were not written before the code; they were read out of it, because the code came first. Setting that description against the workflow that the team needed showed that a substantial number of rules did not match what the hospital wanted, and that some expected actions did not exist at all. Sun, Xu and Su note that access control rules are specific to each application, which makes it hard to get a precise specification for a tool to enforce [8]. Here there was no specification to enforce against: it had to be recovered from the code first.

The consequence is short. A clean scan is evidence about code patterns. It is not evidence that the right people can do the right things, and it should never be recorded as though it were.

6.2.3 Layered Checks, and the Guard That Never Ran

The clearest illustration of layered checking came from a mistake. The file holding the page rules sat outside the folder the framework reads, so it never loaded, and that whole layer had been doing nothing for the life of the project. No additional exposure was demonstrated from the inactive guard, because server-side route checks remained in place; however, this depended on those lower-layer checks being correctly and consistently implemented, not on any property of the missing guard itself. What only emerged later was that the file's own settings excluded those addresses entirely, so even had it loaded, it would never have guarded what it appeared to guard. It read as a control, and was never capable of being one.

Two things follow. Nobody knew the layer was dead until it was found by accident, which is an argument for testing that a control actually runs rather than assuming it does because the code exists. A protection that has never run offers nothing, and it looks exactly like a working one until someone checks. The system survived because the decision was made in one place, which is what OWASP asks for, that access control be implemented once and reused throughout an application [6]. That is not complete: at least eight other places in the code write their own separate copy of the same case-assignment rule instead of calling the shared check. All eight currently agree with each other, but only because each copy has a comment telling a developer to keep it matching the others by hand, nothing in the code forces them to match. This already went wrong once: one of those copies, in the part of the system that lets an anaesthetist open a case, fell out of step with the real rule, and for a time let a surgeon reach a case they had no right to see. It was found and corrected, but nothing stopped it from happening, and nothing stops the same drift from happening again in one of the other seven copies. Rules kept in more than one place drift apart, and this one already has; this is recorded in section 7.3. The same inconsistency shows up in reverse elsewhere in the codebase: the route that manages user accounts checks permission before ever touching the database, so a non-administrator is blocked before the system reveals whether a given user ID is even real, the correct pattern already exists, it simply was not applied everywhere.

The same pattern appeared twice more. In the path traversal finding, an attack failed only because a database check happened to run first; the system was safe, but not for the reason anyone had designed, and changing the order of steps later would have removed that protection without warning. And three background jobs were written, correct, and never ran: nothing on the deployment server started them, and the one call that did reach a job used a method the route did not accept, so it was rejected before anything executed. Each has since taken a different path. One, the job meant to close the gap left by a session that cannot be withdrawn, was later superseded rather than fixed: a live check replaced it rather than the job itself being scheduled, which is a different way of closing a gap than the pattern this section is about, but it means that particular gap is no longer open. Another, the slow-case warning job, was given a working trigger, an in-process timer rather than the originally planned OS-level scheduler, so it now runs as intended, if by a less independent mechanism than first planned. The third, the deadline-closure job, was retired outright rather than fixed or

scheduled.

6.2.4 What This Suggests for Similar Systems

Four points carry beyond this project.

Where a workflow hands responsibility from one group to another, the state of the work belongs in the permission rule alongside the role. Without it, the handover is not enforced at all.

Where authorisation rules come from an organisation rather than from the code, they have to be tested by using the system as each role, because no scanner has access to them.

And where a control matters, it is worth confirming that it runs. Code that exists and code that executes are not the same thing.

Where a system's own rules say what a role should not be able to do, that has to be checked directly against the code, not assumed from a clean scan or from what the screens allow. A role's real reach can extend past its visible interface to routes with no button behind them, invisible to both a scanner and to testing done through the screens; catching that gap takes a third check, one that compares the code to the system's own stated rules rather than to what a scanner or a screen would show.

7. Challenges, Conclusion and Future Work

This chapter covers the difficulties encountered during the security work, the conclusions drawn from it, and the work that remains for later phases.

7.1 Challenges

7.1.1 Technical Challenges

I started with no background and no map. I had not built software before this project, so reading someone else's code and changing it safely were things I learned while doing them. Nor was there a structure to follow. At the beginning my learning was scattered, one topic, then another, in no order. What structure existed came later, from the frameworks in Chapter 4; section 7.1.3 reflects further on how that gap closed.

Security began after development. A teammate wrote the application code first, because I was new to this area and the access rules were not ready, and the team could not wait. That was the right call for a six-month phase, but it set the method: I could not build access control from a written table, because the code already existed. I read the code, wrote down what it allowed each role to do, compared that against the workflow the team needed, and changed the code where they differed. A substantial number of rules did not match what the hospital wanted, and some expected actions did not exist at all.

The domain was entirely new. Electronic medical records, the hospital's own information system, the FHIR standard, and how insurance approval works were all unfamiliar. Before any of us could start building, we spent weeks understanding the process itself, what the hospital already had, what they wanted, and how people actually worked. Access control cannot be designed without understanding the workflow it protects, so learning the domain was a prerequisite rather than background reading.

Evaluation used the time that implementation would have needed. The team spent several months trialling Microsoft Entra ID before it was ruled out; section 7.1.3 describes the decision itself. Federation with the hospital's staff directory was delayed by the time spent on that trial. It is now complete, and section 5.1.2 describes the result.

Almost everything was designed by revision. Access rules were built one at a time, decided, implemented, checked, and sometimes the requirement itself changed once people saw the result. The audit trail had no template at all: I decided what to record, revised it, took it to the IT head, revised again, researched, and revised once more. Verification behaved the same way, since each time I finished checking, a teammate found something that sent the work back round. The gaps recorded in Chapter 6 reflect a design arrived at by iteration rather than one worked out in advance.

7.1.2 Stakeholder Challenges

The workflow had to be understood before it could be secured. Who may do what depends on which step a case has reached, so the flow had to be settled before the rules enforcing it could be written. The flow follows the hospital's existing paper procedure, so I began with the paper, reading the forms, following how a file moved between departments, and learning the terms doctors and hospital staff use for what those forms contain. None of that vocabulary was familiar. The flow was not mine to settle either: the Coordinator and Finance teams had to agree to it between them, over a series of meetings, and each change to it changed the rules enforcing it. Section 4.1 describes the weekly cycle this ran through; what follows here is the difficulty of that process, not a separate account of it.

Even now, with the flow decided and built, it is not certain that it matches what those teams expect in daily use.

Some findings were questions rather than defects. Whether the FHIR server is needed at all is a decision for the hospital, not for me. Whether the Coordinator should keep any ability to change prices after Finance has confirmed them was also such a question; it was raised with the hospital's team and settled during this work, in favour of removing that ability rather than keeping it, described in section 5.1.2. Questions of this kind took longer to settle than code fixes. Changes of that kind also affected my teammates, locking a form or restricting an action alters what they are building against, so they were raised before being applied rather than afterwards.

7.1.3 Personal and Organisational Reflection

Individual

On the first day of this placement, I knew almost nothing relevant to the work ahead: I could not read code, I was unfamiliar with the application's interface, the framework it was built on, or how a hospital actually operates, let alone the specific GOP workflow or the vocabulary staff use for it. What changed this was not becoming a confident programmer myself, but a steady sequence of real conversations, with Finance (Bong Kimheng), with doctors (Dr. Denny, Dr. Visoth, Dr. Justyna), with the Coordinator team (Mrs. Debbie, Nurse Riza), and with the IT team (Bong Udom, Bong Chantha, Bong Savda), alongside guidance from my supervisor. My teammate Rith's own work also shaped how I learned: he built the application's interface and workflow, and once that existed, I could adapt and build my own work against something real rather than something abstract. Data and how it was queried remained genuinely difficult for me; there were times my supervisor, Dr. Elias, would discuss it and I still did not fully follow. But seeing Rith's work running, the actual result, was often what let me understand what a "dream" we had discussed in a meeting actually looked like once it existed, more than the discussion itself ever could. I learned to direct AI tooling to investigate and implement, to check specific things myself where I could, such as configuration files for default credentials, and to verify what the tooling reported against the running system rather than accepting it on trust. That skill, knowing what to check, recognising a wrong answer, and taking responsibility for the final decision, is what actually grew over six months, more than fluency in writing code by hand.

Confidence was something I had to build, not something I started with. On my first days, I was genuinely afraid, across every team I worked with, doctors, Finance, the Coordinator team, IT, my supervisor, and Dr. Ebner, Intercare's Deputy CEO, alike, that I would use the wrong word, explain something poorly, or come across as rude simply by not knowing how to phrase a question. That fear turned out to be unfounded. Rather than judging what I did not yet know, everyone who contributed and participated in this project treated me and my teammates as part of the group, almost like younger siblings, and created an environment where I could ask questions and share half-formed ideas honestly, without it ever tipping into disrespect. That sense of closeness, being able to speak freely while still being taken seriously, is what actually let the weekly stakeholder conversations described in section 4.1 work. I would not have been able to ask any of the teams, doctors, Finance, the Coordinator team, or IT, to correct my assumptions about the workflow, or to speak plainly with my supervisor. Much of that confidence came directly from all of their support.

Team

Within the three-person team, this growth did not start smoothly. Rith, Chamrath, and I were not initially close, and there was an early period of awkwardness where we did not yet know how to talk to one another comfortably. Rith was the one who consistently kept the team connected and made the work feel less tense.

On the security side specifically, I started from genuinely knowing nothing, not even how to build an RBAC and ABAC table, and my first attempt to bring the team into that work was to have AI generate a document and simply share the file for them to read. That did not work: neither Chamrath nor Rith had the background to make sense of it on their own, and without a shared conversation behind it, the team pulled in different directions rather than together, leading to some real disagreement and misunderstanding.

What changed this was replacing the file-sharing with direct conversation. Instead of sending something off for the team to interpret alone, I started talking through the access-control questions with them directly, explaining what I was trying to figure out and asking what would actually work for how they were building the application. That shift took real effort at first. I did not have the words ready, and these conversations took time. But over many repeated discussions, the team reached a point where we understood what each person needed and what a workable outcome looked like for all three of us, including ways of expressing an idea that I could not have put into words earlier in the placement. From that point on, the communication, whether in person, by chat, or by message, became smooth, and my security questions stopped being something delivered to the team and became something worked out with them.

My contribution to the team's design conversations often took the shape of a "what if" question, which I would raise and the team would then discuss together. Two access-control rules described in Chapter 5 came about this way. The first concerned when a doctor's ability to edit a case should end: the original build let a doctor keep editing indefinitely, and I raised the question of what if that ability ended once the case reached Finance's queue, since Finance was meant to be working from a figure the doctors had already committed to. The second concerned what happened when a case was picked up: originally, once an anaesthetist opened an unassigned case it became permanently theirs, with no way to release it back. I raised the same kind of question there too, what if a doctor could hand a case back rather than being stuck with it. Both questions led to real discussion with the team about what should happen, and both are now working rules in the deployed system.

Not every hard choice in this project was about making something secure versus easy for a doctor or Finance to use. Some were about making something secure versus easy for our own team to build. The clearest example was deciding how users would sign in. The original plan was Entra ID, so staff could use the same Microsoft login they already had, without needing to remember another password. When that plan fell through, we first considered building a simple login inside the application itself, since that would have been the easiest for us to set up. But we decided against it: staff would have needed yet another separate password just for this one system, and it was less secure than the alternatives, along with a few other concerns the team raised. I went back and researched further, and found Keycloak. It fit what we actually needed, but it came with a real cost: none of us had used it before, and it took much more setup work than the simpler option would have. After more discussion, the team agreed it was worth the extra effort, and that decision became the login system described in Chapter 5.

Organisation

Intercare gained more than a working application from this project. Before this work, the rules governing who could do what existed only as unwritten habit in the paper process itself, understood by whoever had always done that part of the job, and once the system was built, that same logic lived scattered inside the code, understood only by whoever had written it. It now exists as a reviewed access control rules document, built and revised with the team over several rounds, that anyone can check without reading the code or asking the person who wrote it. The hospital also now has a real audit trail, a tamper-proof record of who did what and when, which did not exist before this work at all, on paper or otherwise. And rather than leaving open issues implicit or undocumented, the future work described in section 7.3 lists exactly what remains, with a priority and an owner against each item, so the hospital's IT team knows what is still open without having to rediscover it themselves.

More broadly, this project was also where the hospital, and the team, learned what it actually takes to turn a rule that existed only as paper habit into a rule the software enforces. On paper, the Coordinator's role in the workflow, or the point at which a doctor should stop being able to edit a case, were things everyone simply understood and followed by habit, never written down as a rule anywhere. Turning each of those into a software rule meant making every implicit assumption explicit, deciding exactly when it should apply, and testing that the system actually behaved that way. That process, more than any single technical control, is what this project leaves behind for Intercare to build on.

Wider Practice

If I had to give advice to another student team starting a similar project, two things stand out. The first is that communication matters more than almost anything else. Most of the delay in this project did not come from not knowing the workflow or the technical background. It came from things I did not dare to say to the other party.

When I first realised I did not know how to read code, I kept that to myself instead of telling my supervisor, Dr. Elias, directly, and carried that self-doubt for months before finally admitting it. His response was not what I feared; instead, he worked hard with me to find a way forward. When I did not know where to start on the workflow, I hesitated to ask, held back by wanting things to be perfect before I even began. It was my mentor, Loukru Makara, who told me to start first and improve along the way, rather than waiting until I felt ready. I also reached out to Bong Udom, Bong Chantha, and Bong Savda, who helped me find at least a starting point, even a small one. With my own teammates, when something needed their feedback or a decision, I sometimes avoided raising it, and the work turned into something I was quietly handling by myself rather than something we solved together. Once I actually raised it with them, they simply took on their part of the responsibility, because it was theirs to carry too. Speaking first and letting the conversation follow worked far better than trying to have the whole answer worked out before saying anything.

The second is to know your limit and accept it honestly. If you only really know A and B, admitting that is enough; there is no need to also claim you know C, or to tell people you can do everything, just to avoid looking less capable. Pretending removes the pressure to actually learn what you are missing, and it closes off the space you need to grow. Someone who never honestly admits their own limit and their own worth never really grows past it. Accepting where you actually stand is what makes it possible to ask for help, learn the next thing, and genuinely improve, rather than staying stuck pretending you already have. If there is one thing I would say underneath both of these, it is: do not run for perfection, run for progress.

7.2 Conclusion

This project secured a system that automates the Guarantee of Payment process at a Cambodian hospital. An identity provider was deployed on the hospital's own servers, access control was built across forty-two defined actions for five roles, an append-only audit trail was added, and the result was checked by two methods before deployment.

The clearest result came from comparing those methods. Static analysis applied 257 rules across 356 files and returned fifteen findings, none about access control. Role-by-role testing of the same system in the same period produced forty-seven findings, twenty-three of them access control. The scanner was not badly configured: access rules here come from how the hospital divides responsibility, not from anything visible in the shape of the code, and a tool cannot check a rule it was never given.

The second result concerns what those rules depend on. A permission is decided by three things together, the user's role, whether the case is theirs, and how far the case has progressed. The third is easiest to miss, because it changes the answer for a user whose role and assignment have not changed: a surgeon becomes read-only when the anaesthetist finishes and the case moves to Finance. That step is what enforces in software the separation the hospital already practised on paper.

The limits are stated rather than implied. There is no automated test suite, so verification was manual and happened once. Several controls a mature deployment would need remain incomplete rather than partially built: testing under heavy load, multi-factor authentication, encryption on two internal connections, alerting on the audit and activity logs, and a data-retention policy, each recorded with an owner in section 7.3. The system was served without encryption at the time of assessment, and two of its internal connections still are. Everything reported in this thesis is self-assessment, by the person who built the controls; an independent penetration test has since been carried out by Canadia Bank (IT security department) and the hospital's IT Security Division, but its findings are confidential and not reproduced in this thesis, so this thesis cannot say whether it confirmed or contradicted what is reported here. And the final review found controls that had never run at all: a page-rule file the framework never loaded, and three background jobs nothing on the server started. Each has since taken a different path, superseded, given a working trigger, or retired outright, described in section 5.4, but none of the three were fixed by the simple means first planned. That combination is the plainest evidence that a system checked once is not a system that stays checked.

A third result concerns where these rules actually came from. Writing them down correctly took more than applying a framework; it took sitting with Finance, the Coordinator team, doctors, and IT, week after week, until the team genuinely understood how responsibility moved between departments on paper, described in section 4.1. Only then could that understanding become a rule the code could enforce. Section 7.1.3 reflects further on what this placement changed beyond the system itself, for me individually, for the team, and for the hospital.

What this project produced is therefore not a secure system but a documented one: rules that can be checked, a record of what was found, a list of what remains with an owner against each item, and an account of how organisational responsibility was translated into software in the first place.

7.3 Future Work

Twenty-three items remain. For a reader who wants only the highest operational priorities before the full table: complete internal TLS between Keycloak, the application, and Active Directory; move Keycloak out of development mode; close the System Administrator write-access gap on clinical and cost data; add multi-factor authentication; automate security scanning in the build pipeline; and introduce monitoring and alerting on the audit and activity logs. Some items are not mine to decide. Setting the audit retention period, or deciding whether the disk should be encrypted, are choices for the hospital. Some need another team, such as the server administrator or Canada Bank (IT security department), since they sit outside the application itself. The rest were within reach but ran out of time, and are left for the next phase.

Table 14: Work remaining, by priority

#	Item	Category	Priority	Who can carry it
1	Encrypt the Keycloak↔application and Keycloak↔AD connections	Server config	High	Server administrator
2	Move Keycloak out of development mode in the production deployment profile, which its own documentation advises against	Server config	High	Server administrator
3	Add multi-factor authentication	Development	High	Security
4	Close System Administrator's write access to clinical and cost data on two routes, since the design says this role should be view-only there	Development	High	Security
5	Hide raw error messages from users in production (ASVS F4)	Development	Medium	Security
6	Ask for password again before finalising a case or sending it to the insurer (ASVS F6)	Development	Medium	Security
7	Send an alert when someone fails to log in or access something too many times	Development	Medium	Security
8	Record device/network address in audit entries	Development	Medium	Security
9	Add a way to force-log-out one specific session, without disabling the whole account	Development	Medium	Security
10	Check if the server can handle heavy use and stay running (DoS / availability)	Server config	Medium	Canada Bank (IT security department) (reviewed), Hospital IT

11	Add automated dependency and code scanning to the build pipeline, so a fixed vulnerability does not silently reappear in a later release without anyone noticing	Server config	Medium	Security
12	Write down when someone is blocked for having no role, since this is not recorded anywhere yet	Design	Medium	Security
13	Add city, country, and gender to the activity log's redaction list, since these three fields are not yet covered and would still be stored in full if a request happened to carry them	Development	Low	Security
14	Validate that the audit log's recorded address actually came through a trusted proxy, rather than reading a header a client can set itself	Development	Medium	Security
15	Extend the audit trigger so a clinical-form edit made after the surgeon has verified, but before the case reaches Finance, is also recorded	Development	Low	Security
16	Combine the case-assignment rule into one place (8 separate copies currently exist)	Design	Medium	Security
17	Re-assess the system against a newer version of ASVS (5.0), since 4.0.3 and OWASP Top 10:2021 are both now several years old	Assessment	Medium	Security
18	Confirm whether Canada Bank's container review specifically followed the CIS Docker Benchmark, or complete that comparison separately if not	Assessment	Low	Canada Bank (IT security department) (reviewed), Hospital IT
19	Confirm whether Canada Bank's penetration test specifically followed the OWASP Testing Guide, or complete that comparison separately if not	Assessment	Low	Canada Bank (IT security department) (reviewed), Hospital IT
20	Make the Actor column match the Description text after someone renames	Design	Low	Security
21	Give the same "not allowed" message whether a case is real or fake	Design	Low	Security
22	Fully remove the developer sign-in shortcut's code, rather than leaving it present and flag-gated, so it cannot be reintroduced by a future configuration change	Development	Medium	Security
23	Make the browser idle timer account for time elapsed while a tab was closed, rather than resetting to zero on reopen, since closing the tab currently gives no protection against a different person opening the browser soon after	Development	Medium	Security

References

[1] Intercare Hospital. [Online]. Available: <https://intercarehospital.com/>. [Accessed: Jul. 28, 2026].

- [2] HL7 International, "Security," HL7 FHIR Release 5. [Online]. Available: <https://hl7.org/fhir/security.html>. [Accessed: Jul. 28, 2026].
- [3] V. C. Hu, D. Ferraiolo, R. Kuhn, A. Schnitzer, K. Sandlin, R. Miller, and K. Scarfone, "Guide to Attribute Based Access Control (ABAC) Definition and Considerations," National Institute of Standards and Technology, NIST Special Publication 800-162, Jan. 2014. [Online]. Available: <https://nvlpubs.nist.gov/nistpubs/specialpublications/nist.sp.800-162.pdf>. [Accessed: Jul. 28, 2026].
- [4] National Bank of Cambodia, "Technology and Cyber Risk Management Guidelines," Jan. 2026. [Online]. Available: https://www.nbc.gov.kh/download_files/publication/itguideline_eng/National_Bank_of_Cambodia_TCRMG_2026.pdf. [Accessed: Jul. 28, 2026].
- [5] National Institute of Standards and Technology, "The NIST Cybersecurity Framework (CSF) 2.0," NIST Cybersecurity White Paper NIST CSWP 29, Feb. 2024. [Online]. Available: <https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>. [Accessed: Jul. 28, 2026].
- [6] Open Worldwide Application Security Project, "OWASP Top 10:2021." [Online]. Available: <https://owasp.org/Top10/2021/>. [Accessed: Jul. 28, 2026].
- [7] Microsoft, "The STRIDE Threat Model," Microsoft Learn. [Online]. Available: [https://learn.microsoft.com/en-us/previous-versions/commerce-server/ee823878\(v=cs.20\)](https://learn.microsoft.com/en-us/previous-versions/commerce-server/ee823878(v=cs.20)). [Accessed: Jul. 28, 2026].
- [8] F. Sun, L. Xu, and Z. Su, "Static Detection of Access Control Vulnerabilities in Web Applications," in Proc. 20th USENIX Security Symposium, San Francisco, CA, 2011. [Online]. Available: <https://www.usenix.org/conference/usenix-security-11/static-detection-access-control-vulnerabilities-web-applications>. [Accessed: Jul. 29, 2026].
- [9] A. Anas, S. Elgamal, and B. Youssef, "Survey on detecting and preventing web application broken access control attacks," International Journal of Electrical and Computer Engineering (IJECE), vol. 14, no. 1, pp. 772–781, Feb. 2024, doi: 10.11591/ijece.v14i1.pp772-781.
- [10] J. Zhu, B. Chu, H. Lipford, and T. Thomas, "Mitigating Access Control Vulnerabilities through Interactive Static Analysis," in Proc. 20th ACM Symposium on Access Control Models and Technologies, Jun. 2015, doi: 10.1145/2752952.2752976.
- [11] U. N. Cobrado, S. Sharief, N. G. Regahal, E. Zepka, M. Mamauag, and L. C. Velasco, "Access control solutions in electronic health record systems: A systematic review," Informatics in Medicine Unlocked, vol. 49, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352914824001084>. [Accessed: Jul. 29, 2026].
- [12] H. F. Atlam and Y. Yang, "Enhancing Healthcare Security: A Unified RBAC and ABAC Risk-Aware Access Control Approach," Future Internet, vol. 17, no. 6, art. 262, Jun. 2025, doi: 10.3390/fi17060262.
- [13] Open Worldwide Application Security Project, "OWASP Top 10:2025." [Online]. Available: <https://owasp.org/Top10/2025/>. [Accessed: Jul. 28, 2026].
- [14] Open Worldwide Application Security Project, "OWASP Application Security Verification Standard (ASVS)." [Online]. Available: <https://owasp.org/www-project-application-security-verification-standard/>. [Accessed: Sep. 13, 2026].
- [15] U.S. Department of Health and Human Services, "The Security Rule." [Online]. Available: <https://www.hhs.gov/hipaa/for-professionals/security/index.html>. [Accessed: Sep. 15, 2026].
- [16] International Organization for Standardization, "ISO/IEC 27001:2022, Information security management systems." [Online]. Available: <https://www.iso.org/standard/27001>. [Accessed: Sep. 15, 2026].